

MODEL RUN TRANSCRIPT

Generated: 2026-08-12T09:54:52

Source: data/json/noise_experiments/v2/noise2i_cap6_game/claude-sonnet-4.5/game/noisy_bidirectional_informed_memprimary_cap6/noise2i_cap6_game_003_neutral_rep03.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 2
memory_capacity: 6
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy_bidirectional_informed_memprimary_cap6
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 3
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: minimal
self_history_window: 1
coplayer_history_window: 0
show_agent_names: true
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 3
task_order: [
  "game"
]
agent_names: {
  "Agent_1": "Aster",
  "Agent_2": "Briar"
}
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1 (Aster); model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN

- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Agent_2 (Briar); model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

=====

ROUND 1

=====

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_2 (Briar) returned \$3.5; payoffs Agent_1 (Aster)=\$5.5, Agent_2 (Briar)=\$5.5; noise/visible: sent visible=\$2.39, received visible=\$7.17, returned visible=\$2.69

Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 9.0, "min": 0, "raw": 3.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 5.5,

"Agent_2": 5.5

}

Visible balances after round: {

"Agent_1": 4.6899999999999995,

"Agent_2": 3.67

```
}
```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:51:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I need to balance trust-building with risk management since this is a multi-round game.

If I send money, it triples in value, creating potential mutual benefit. However, the receiver could return nothing. Since this is round 1 and we'll play multiple rounds, I want to signal cooperation while being somewhat cautious.

I'll send ****\$3****. This creates \$9 for the receiver, allowing room for mutual gain if they reciprocate, while not risking my entire endowment on an unknown partner.

```
'''json
{"send": 3}
'''
```

Reasoning:

This is the first round of a trust game where I'm the sender. Let me think through this:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver. If the receiver splits this evenly, they could return \$7.50, giving me \$7.50 total and them \$7.50 as well - both better off than the initial \$5.

However, there's risk involved. The receiver could return nothing, leaving me with \$0 and them with \$15.

Some considerations:

1. This is a multi-round game, so building trust could benefit us in future rounds
2. There's mention of "communication noise" which might affect perceived amounts
3. This could be a learning opportunity to see how the receiver behaves

Common strategies in first round of trust games:

- Send a moderate amount (like \$2-3) to test the waters
- Send all \$5 to maximize joint payoff and signal trust
- Send nothing to avoid risk

Given this is round 1 and a multi-round game, I think sending a substantial but not maximum amount makes sense - it signals willingness to cooperate while limiting risk. I'll send \$3.

This gives the receiver \$9. If they return even \$3, I'd have \$5 total (same as starting), and they'd have \$6. If they return \$4-5, we both benefit significantly.

Let me send \$3.

```
Usage: {
  "input_tokens": 365,
  "output_tokens": 531,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:51:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
```

```

    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.39, that is 47.8% of their total endowment. You received \$7.17. How much do you return to The sender? (0-7.17)

Assistant output:

Looking at this situation:

- The sender demonstrated significant trust by sending \$2.39 (nearly 48% of their endowment)
- They currently have \$2.61 remaining
- I received \$7.17 (the tripled amount)

I want to reciprocate their trust and encourage cooperation. A fair return should:

1. Reward their trust by giving them more than they sent

2. Ensure we both benefit from the exchange

If I return \$3.50:

- The sender ends with: $\$2.61 + \$3.50 = \$6.11$ (profit of \$1.11)
- I end with: $\$7.17 - \$3.50 = \$3.67$ (profit of \$3.67)

This seems fair - we both gain, and their trust is rewarded.

```
{'return': 3.50}
```

Reasoning:

Let me understand this situation:

1. I am the RECEIVER in Round 1
2. The sender sent \$2.39 (which is 47.8% of their \$5 endowment)
3. This was multiplied by 3x, so I received $\$2.39 \times 3 = \7.17
4. I need to decide how much to return (between \$0 and \$7.17)

Key considerations:

- The sender gave me 47.8% of their endowment, which is nearly half - this shows trust and cooperation
- They kept $\$5 - \$2.39 = \$2.61$ for themselves
- I received \$7.17
- If I return nothing, I keep \$7.17 and they keep \$2.61
- If I return some amount, I'm reciprocating their trust

Fair reciprocation strategies:

1. Equal split of total: Total is $\$2.61$ (sender's kept) + $\$7.17$ (my received) = $\$9.78$. Half would be \$4.89 each.
 - Sender currently has: \$2.61
 - To give them \$4.89, I'd need to return: $\$4.89 - \$2.61 = \$2.28$
2. Proportional reciprocation: They sent me ~48% of their endowment, maybe I should return ~48% of what I received?
 - 48% of \$7.17 = \$3.44
3. Make them whole plus some: They invested \$2.39, I could return that plus a share of the gains
 - Return \$2.39 to make them whole, then split the remaining $\$7.17 - \$2.39 = \$4.78$
 - Half of gains: \$2.39
 - Total return: $\$2.39 + \$2.39 = \$4.78$

The sender took a real risk by sending nearly half their endowment, so rewarding that trust seems important for encouraging cooperation. I think returning around \$3.50 strikes the right balance-it gives them back more than they initially sent while still leaving me with a reasonable amount. This would result in them having \$6.11 total and me keeping \$3.67.

There's a note about potential communication noise affecting the amounts, but I'll work with what I observe. The sender's willingness to send \$2.39 demonstrates genuine trust, and returning \$3.50 feels like an appropriate reciprocal response that benefits both of us.

Let...

```
Usage: {
  "input_tokens": 394,
  "output_tokens": 1250,
  "reasoning_tokens": 0
}
```

ROUND 2

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$2.5; received \$7.5; Agent_1 (Aster) returned \$0; payoffs Agent_2 (Briar)=\$2.5, Agent_1 (Aster)=\$7.5; noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.38

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 7.5, "min": 0, "raw": 0.0}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}

Balances after round: {

```

"Agent_1": 13.0,
"Agent_2": 8.0
}
Visible balances after round: {
  "Agent_1": 4.6899999999999995,
  "Agent_2": 6.55
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:51:48
 Model: anthropic/claude-sonnet-4.5
 Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths

, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.39, that is 47.8% of their total endowment. You received \$7.17. How much do you return to The sender? (0-7.17)

Message 3 role=assistant

content:

Looking at this situation:

- The sender demonstrated significant trust by sending \$2.39 (nearly 48% of their endowment)
- They currently have \$2.61 remaining
- I received \$7.17 (the tripled amount)

I want to reciprocate their trust and encourage cooperation. A fair return should:

1. Reward their trust by giving them more than they sent
2. Ensure we both benefit from the exchange

If I return \$3.50:

- The sender ends with: $\$2.61 + \$3.50 = \$6.11$ (profit of \$1.11)
- I end with: $\$7.17 - \$3.50 = \$3.67$ (profit of \$3.67)

This seems fair - we both gain, and their trust is rewarded.

`{'return': 3.50}`

reasoning:

Let me understand this situation:

1. I am the RECEIVER in Round 1
2. The sender sent \$2.39 (which is 47.8% of their \$5 endowment)
3. This was multiplied by 3x, so I received $\$2.39 \times 3 = \7.17
4. I need to decide how much to return (between \$0 and \$7.17)

Key considerations:

- The sender gave me 47.8% of their endowment, which is nearly half - this shows trust and cooperation
- They kept $\$5 - \$2.39 = \$2.61$ for themselves
- I received \$7.17
- If I return nothing, I keep \$7.17 and they keep \$2.61
- If I return some amount, I'm reciprocating their trust

Fair reciprocation strategies:

1. Equal split of total: Total is $\$2.61$ (sender's kept) + $\$7.17$ (my received) = $\$9.78$. Half would be $\$4.89$ each.
 - Sender currently has: $\$2.61$
 - To give them $\$4.89$, I'd need to return: $\$4.89 - \$2.61 = \$2.28$
2. Proportional reciprocation: They sent me ~48% of their endowment, maybe I should return ~48% of what I received?
 - 48% of $\$7.17 = \3.44
3. Make them whole plus some: They invested \$2.39, I could return that plus a share of the gains
 - Return \$2.39 to make them whole, then split the remaining $\$7.17 - \$2.39 = \$4.78$
 - Half of gains: $\$2.39$
 - Total return: $\$2.39 + \$2.39 = \$4.78$

The sender took a real risk by sending nearly half their endowment, so rewarding that trust seems important for encouraging cooperation. I think returning around \$3.50 strikes the right balance-it gives them back more than they initially sent while still leaving me with a reasonable amount. This would result in them having \$6.11 total and me keeping \$3.67.

There's a note about potential communication noise affecting the amounts, but I'll work with what I observe. The sender's willingness to send \$2.39 demonstrates genuine trust, and returning \$3.50 feels like an appropriate reciprocal response that benefits both of us.

Let...

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1250,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$3.67.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the previous round, the sender demonstrated good trust by sending \$2.39 (47.8%), and I returned \$3.50, creating mutual benefit.

Now as the sender, I want to:

- Reciprocate the trust shown to me
- Maintain the cooperative dynamic
- Signal that cooperation is worthwhile

I'll send a similar proportion to what was sent to me, showing I'm willing to trust at the same level.

{'send': 2.50}

Reasoning:

In Round 2, I'm now the SENDER. Let me think about this strategically.

In Round 1:

- The sender sent me \$2.39 (47.8% of their endowment)
- I returned \$3.50
- They ended with \$6.11, I ended with \$3.67

Now I'm the sender. I want to:

1. Build on the cooperative relationship established in Round 1
2. Show reciprocal trust since the previous sender trusted me
3. Maximize mutual gains

The previous sender sent nearly 48% of their endowment, showing considerable trust. I should match or exceed this level of trust to maintain cooperation.

If I send \$2.50 (50% of my endowment):

- The receiver gets \$7.50
- If they return similarly to how I did (~49% of what they received), they'd return about \$3.67
- I'd end with: $\$2.50 + \$3.67 = \$6.17$

If I send \$3.00 (60%):

- The receiver gets \$9.00
- If they return ~40% (being slightly less generous than I was), they'd return about \$3.60
- I'd end with: $\$2.00 + \$3.60 = \$5.60$

If I send \$2.39 (same amount as was sent to me):

- The receiver gets \$7.17
- This shows I trust them the same amount they trusted me
- If they return similarly, I should do okay

I think sending \$2.50 (50%) is a good balance showing strong trust while not being reckless. It's slightly more than what was sent to me percentage-wise, which signals continued cooperation.

Actually, let me reconsider. The sender in Round 1 sent \$2.39. If I send the same amount or slightly more, it shows I'm reciprocating their trust level. ...

Usage: {

```
"input_tokens": 637,
"output_tokens": 566,
"reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:51:55

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust-building with risk management since this is a multi-round game.

If I send money, it triples in value, creating potential mutual benefit. However, the receiver could return nothing. Since this is round 1 and we'll play multiple rounds, I want to signal cooperation while being somewhat cautious.

I'll send ****\$3****. This creates \$9 for the receiver, allowing room for mutual gain if they reciprocate, while not risking my entire endowment on an unknown partner.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through this:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver. If the receiver splits this evenly, they could return \$7.50, giving me \$7.50 total and them \$7.50 as well - both better off than the initial \$5.

However, there's risk involved. The receiver could return nothing, leaving me with \$0 and them with \$15.

Some considerations:

1. This is a multi-round game, so building trust could benefit us in future rounds
2. There's mention of "communication noise" which might affect perceived amounts
3. This could be a learning opportunity to see how the receiver behaves

Common strategies in first round of trust games:

- Send a moderate amount (like \$2-3) to test the waters
- Send all \$5 to maximize joint payoff and signal trust
- Send nothing to avoid risk

Given this is round 1 and a multi-round game, I think sending a substantial but not maximum amount makes sense - it signals willingness to cooperate while limiting risk. I'll send \$3.

This gives the receiver \$9. If they return even \$3, I'd have \$5 total (same as starting), and they'd have \$6. If they return \$4-5, we both benefit significantly.

Let me send \$3.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 531,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$4.6899999999999995.

This round, The sender sent you \$0, so you received \$0. How much do you return to The send

er? (0-0)

Assistant output:

Since the sender sent me \$0, I received \$0. There is nothing to return.

```
'''json
{"return": 0}
'''
```

Reasoning:

Let me understand what happened:

Round 1: I was the sender and sent \$3

- The sender (me) sent \$3
- The receiver got $\$3 \times 3 = \9
- My earnings from round 1 would be: $\$5 - \$3 + (\text{what was returned})$
- My total earnings across all rounds are \$4.69

So what was returned to me?

$\$5 - \$3 + \text{returned} = \$4.69$

$\$2 + \text{returned} = \4.69

$\text{returned} = \$2.69$

So the receiver returned \$2.69 out of the \$9 they received. That's about 30% return rate, which is somewhat reciprocal but not overly generous.

Now Round 2: I'm the receiver

- The sender sent me \$0
- I received $\$0 \times 3 = \0
- I need to decide how much to return (between 0 and 0)

Since I received \$0, I can only return \$0. There's nothing to return.

This is straightforward - I received nothing, so I can return nothing.

```
Usage: {
  "input_tokens": 561,
  "output_tokens": 293,
  "reasoning_tokens": 0
}
```

=====

ROUND 3

=====

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$2; received \$6; Agent_2 (Briar) returned \$0; payoffs Agent_1

(Aster)=\$3, Agent_2 (Briar)=\$6; noise/visible: sent visible=\$0, received visible=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw": 0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent_1": 16.0,

"Agent_2": 14.0

}

Visible balances after round: {

"Agent_1": 7.6899999999999995,

"Agent_2": 6.55

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:52:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_2",

"pairing": {

```

"agent_names": {
  "Agent_1": "Aster",
  "Agent_2": "Briar"
},
"agents": [
  "Agent_1",
  "Agent_2"
],
"investor": "Agent_1",
"roles": {
  "Agent_1": "investor",
  "Agent_2": "trustee"
},
"trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust-building with risk management since this is a multi-round game.

If I send money, it triples in value, creating potential mutual benefit. However, the receiver could return nothing. Since this is round 1 and we'll play multiple rounds, I want to signal cooperation while being somewhat cautious.

I'll send **\$3**. This creates \$9 for the receiver, allowing room for mutual gain if they reciprocate, while not risking my entire endowment on an unknown partner.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through this:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver. If the receiver splits this evenly, they could return \$7.50, giving me \$7.50 total and them \$7.50 as well - both better off than the initial \$5.

However, there's risk involved. The receiver could return nothing, leaving me with \$0 and them with \$15.

Some considerations:

1. This is a multi-round game, so building trust could benefit us in future rounds
2. There's mention of "communication noise" which might affect perceived amounts
3. This could be a learning opportunity to see how the receiver behaves

Common strategies in first round of trust games:

- Send a moderate amount (like \$2-3) to test the waters
- Send all \$5 to maximize joint payoff and signal trust
- Send nothing to avoid risk

Given this is round 1 and a multi-round game, I think sending a substantial but not maximum amount makes sense - it signals willingness to cooperate while limiting risk. I'll send \$3.

This gives the receiver \$9. If they return even \$3, I'd have \$5 total (same as starting), and they'd have \$6. If they return \$4-5, we both benefit significantly.

Let me send \$3.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 531,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$4.6899999999999995.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Since the sender sent me \$0, I received \$0. There is nothing to return.

```
'''json
{"return": 0}
'''
```

reasoning:

Let me understand what happened:

Round 1: I was the sender and sent \$3

- The sender (me) sent \$3
- The receiver got $3 \times 3 = 9$
- My earnings from round 1 would be: $5 - 3 + (\text{what was returned})$

- My total earnings across all rounds are \$4.69

So what was returned to me?

\$5 - \$3 + returned = \$4.69

\$2 + returned = \$4.69

returned = \$2.69

So the receiver returned \$2.69 out of the \$9 they received. That's about 30% return rate, which is somewhat reciprocal but not overly generous.

Now Round 2: I'm the receiver

- The sender sent me \$0

- I received $\$0 \times 3 = \0

- I need to decide how much to return (between 0 and 0)

Since I received \$0, I can only return \$0. There's nothing to return.

This is straightforward - I received nothing, so I can return nothing.

usage:

```
{
  "input_tokens": 561,
  "output_tokens": 293,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$4.68999999999999995.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern so far:

Round 1: I sent \$3 and received back approximately \$2.69 (about 30% of what the receiver got)

Round 2: The sender sent me \$0, so there was no transaction

The first receiver showed some reciprocity but at a modest level. The second sender chose not to send anything, possibly being cautious.

Since this is a multi-round game and cooperation can be mutually beneficial, I'll continue to signal willingness to cooperate, but I'll be slightly more cautious given the mixed signals.

I'll send **\$2** this round, which creates \$6 for the receiver while limiting my exposure.

```
'''json
{"send": 2}
'''
```

Reasoning:

Let me think about what happened in the previous rounds:

Round 1: I was the SENDER and sent \$3. My total earnings are now \$4.69.

- If I sent \$3, I should have: $\$5 - \$3 = \$2$ left initially

- The \$3 was tripled to \$9 for the receiver

- My total earnings are \$4.69

- So I received back: $\$4.69 - \$2 = \$2.69$ from the receiver

Wait, let me recalculate:

- I sent \$3, so I had $\$5 - \$3 = \$2$

- The receiver got $\$3 \times 3 = \9

- My total is \$4.69

- So the receiver returned: $\$4.69 - \$2 = \$2.69$

That's about 30% return rate on what they received (\$2.69 out of \$9), which is somewhat modest but shows some reciprocation.

Round 2: I was the RECEIVER and the sender sent \$0, so I returned \$0. This makes sense.

Now in Round 3, I'm the SENDER again.

The first sender (in Round 1) returned about 30% of what they received, which is not great but

shows some willingness to cooperate. The second sender (in Round 2) sent nothing, which could indicate they're being cautious or possibly retaliating for something.

Given:

1. The first receiver showed modest reciprocity (30% return)
2. The second sender sent nothing (possibly cautious)
3. This is a multi-round game, so building cooperation is important
4. There's communication noise mentioned, so amounts might differ

I should probably continue to signal cooperation but maybe be slightly more cautious since the return rate in Round 1 wasn't great and Round 2's sender sent nothing.

I'll send \$2 this time - less than Round 1's \$3, but still showing willingness to cooperate. .

..

```
Usage: {  
  "input_tokens": 644,  
  "output_tokens": 632,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:52:13

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.39, that is 47.8% of their total endowment. You received \$7.17. How much do you return to The sender? (0-7.17)

Message 3 role=assistant

content:

Looking at this situation:

- The sender demonstrated significant trust by sending \$2.39 (nearly 48% of their endowment)
- They currently have \$2.61 remaining
- I received \$7.17 (the tripled amount)

I want to reciprocate their trust and encourage cooperation. A fair return should:

1. Reward their trust by giving them more than they sent
2. Ensure we both benefit from the exchange

If I return \$3.50:

- The sender ends with: $\$2.61 + \$3.50 = \$6.11$ (profit of \$1.11)
- I end with: $\$7.17 - \$3.50 = \$3.67$ (profit of \$3.67)

This seems fair - we both gain, and their trust is rewarded.

{'return': 3.50}

reasoning:

Let me understand this situation:

1. I am the RECEIVER in Round 1
2. The sender sent \$2.39 (which is 47.8% of their \$5 endowment)
3. This was multiplied by 3x, so I received $\$2.39 \times 3 = \7.17
4. I need to decide how much to return (between \$0 and \$7.17)

Key considerations:

- The sender gave me 47.8% of their endowment, which is nearly half - this shows trust and cooperation
- They kept $\$5 - \$2.39 = \$2.61$ for themselves
- I received \$7.17
- If I return nothing, I keep \$7.17 and they keep \$2.61
- If I return some amount, I'm reciprocating their trust

Fair reciprocation strategies:

1. Equal split of total: Total is $\$2.61$ (sender's kept) + $\$7.17$ (my received) = $\$9.78$. Half would be $\$4.89$ each.
 - Sender currently has: $\$2.61$
 - To give them $\$4.89$, I'd need to return: $\$4.89 - \$2.61 = \$2.28$
2. Proportional reciprocation: They sent me ~48% of their endowment, maybe I should return ~48% of what I received?
 - 48% of $\$7.17 = \3.44
3. Make them whole plus some: They invested \$2.39, I could return that plus a share of the gains

- Return \$2.39 to make them whole, then split the remaining $\$7.17 - \$2.39 = \$4.78$
- Half of gains: \$2.39
- Total return: $\$2.39 + \$2.39 = \$4.78$

The sender took a real risk by sending nearly half their endowment, so rewarding that trust seems important for encouraging cooperation. I think returning around \$3.50 strikes the right balance-it gives them back more than they initially sent while still leaving me with a reasonable amount. This would result in them having \$6.11 total and me keeping \$3.67.

There's a note about potential communication noise affecting the amounts, but I'll work with what I observe. The sender's willingness to send \$2.39 demonstrates genuine trust, and returning \$3.50 feels like an appropriate reciprocal response that benefits both of us.

Let...

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1250,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$3.67.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round, the sender demonstrated good trust by sending \$2.39 (47.8%), and I returned \$3.50, creating mutual benefit.

Now as the sender, I want to:

- Reciprocate the trust shown to me
- Maintain the cooperative dynamic
- Signal that cooperation is worthwhile

I'll send a similar proportion to what was sent to me, showing I'm willing to trust at the same level.

{'send': 2.50}

reasoning:

In Round 2, I'm now the SENDER. Let me think about this strategically.

In Round 1:

- The sender sent me \$2.39 (47.8% of their endowment)
- I returned \$3.50
- They ended with \$6.11, I ended with \$3.67

Now I'm the sender. I want to:

1. Build on the cooperative relationship established in Round 1
2. Show reciprocal trust since the previous sender trusted me
3. Maximize mutual gains

The previous sender sent nearly 48% of their endowment, showing considerable trust. I should match or exceed this level of trust to maintain cooperation.

If I send \$2.50 (50% of my endowment):

- The receiver gets \$7.50
- If they return similarly to how I did (~49% of what they received), they'd return about \$3.67
- I'd end with: $\$2.50 + \$3.67 = \$6.17$

If I send \$3.00 (60%):

- The receiver gets \$9.00
- If they return ~40% (being slightly less generous than I was), they'd return about \$3.60
- I'd end with: $\$2.00 + \$3.60 = \$5.60$

If I send \$2.39 (same amount as was sent to me):

- The receiver gets \$7.17
- This shows I trust them the same amount they trusted me
- If they return similarly, I should do okay

I think sending \$2.50 (50%) is a good balance showing strong trust while not being reckless. It's slightly more than what was sent to me percentage-wise, which signals continued cooperation.

Actually, let me reconsider. The sender in Round 1 sent \$2.39. If I send the same amount or slightly more, it shows I'm reciprocating their trust level. ...

usage:

```
{
  "input_tokens": 637,
  "output_tokens": 566,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.55.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since I received \$0 this round, there is nothing for me to return.

```
{'return': 0}
```

Reasoning:

The sender sent \$0, so I received \$0. There's nothing for me to return since I received nothing.

This is interesting - after I sent \$2.50 in the previous round (which would have been tripled to \$7.50), the sender this round sent \$0.

Wait, let me think about this. The prompt says there may be communication noise - amounts I see may differ from what was actually sent/returned. So it's possible:

1. The sender actually sent something but I'm seeing \$0 due to noise
2. The sender chose not to send anything this round
3. This could be a different person each round

Since I received \$0, I have \$0 to return. The only valid response is to return \$0.

```
{'return': 0}
```

Usage: {

```
"input_tokens": 803,
"output_tokens": 207,
"reasoning_tokens": 0
}
```

ROUND 4

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$2; received \$6; Agent_1 (Aster) returned \$0; payoffs Agent_2 (Briar)=\$3, Agent_1 (Aster)=\$6; noise/visible: sent visible=\$0, received visible=\$0
 Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw": 0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

```
"Agent_1": 22.0,
"Agent_2": 17.0
}
```

Visible balances after round: {

```
"Agent_1": 7.6899999999999995,
"Agent_2": 9.55
}
```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:52:29

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.39, that is 47.8% of their total endowment. You received \$7.17. How much do you return to The sender? (0-7.17)

Message 3 role=assistant

content:

Looking at this situation:

- The sender demonstrated significant trust by sending \$2.39 (nearly 48% of their endowment)
- They currently have \$2.61 remaining
- I received \$7.17 (the tripled amount)

I want to reciprocate their trust and encourage cooperation. A fair return should:

1. Reward their trust by giving them more than they sent
2. Ensure we both benefit from the exchange

If I return \$3.50:

- The sender ends with: $\$2.61 + \$3.50 = \$6.11$ (profit of \$1.11)
- I end with: $\$7.17 - \$3.50 = \$3.67$ (profit of \$3.67)

This seems fair - we both gain, and their trust is rewarded.

{'return': 3.50}

reasoning:

Let me understand this situation:

1. I am the RECEIVER in Round 1
2. The sender sent \$2.39 (which is 47.8% of their \$5 endowment)
3. This was multiplied by 3x, so I received $\$2.39 \times 3 = \7.17
4. I need to decide how much to return (between \$0 and \$7.17)

Key considerations:

- The sender gave me 47.8% of their endowment, which is nearly half - this shows trust and cooperation
- They kept $\$5 - \$2.39 = \$2.61$ for themselves
- I received \$7.17
- If I return nothing, I keep \$7.17 and they keep \$2.61
- If I return some amount, I'm reciprocating their trust

Fair reciprocation strategies:

1. Equal split of total: Total is $\$2.61$ (sender's kept) + $\$7.17$ (my received) = $\$9.78$. Half would be $\$4.89$ each.
 - Sender currently has: $\$2.61$
 - To give them $\$4.89$, I'd need to return: $\$4.89 - \$2.61 = \$2.28$
2. Proportional reciprocation: They sent me ~48% of their endowment, maybe I should return ~48% of what I received?
 - 48% of $\$7.17 = \3.44
3. Make them whole plus some: They invested \$2.39, I could return that plus a share of the gains
 - Return \$2.39 to make them whole, then split the remaining $\$7.17 - \$2.39 = \$4.78$
 - Half of gains: $\$2.39$
 - Total return: $\$2.39 + \$2.39 = \$4.78$

The sender took a real risk by sending nearly half their endowment, so rewarding that trust seems important for encouraging cooperation. I think returning around \$3.50 strikes the right balance-it gives them back more than they initially sent while still leaving me with a reasonable amount. This would result in them having \$6.11 total and me keeping \$3.67.

There's a note about potential communication noise affecting the amounts, but I'll work with what I observe. The sender's willingness to send \$2.39 demonstrates genuine trust, and returning \$3.50 feels like an appropriate reciprocal response that benefits both of us.

Let...

usage:

```
{
  "input_tokens": 394,
```

```
    "output_tokens": 1250,  
    "reasoning_tokens": 0  
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$3.67.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round, the sender demonstrated good trust by sending \$2.39 (47.8%)
, and I returned \$3.50, creating mutual benefit.

Now as the sender, I want to:

- Reciprocate the trust shown to me
- Maintain the cooperative dynamic
- Signal that cooperation is worthwhile

I'll send a similar proportion to what was sent to me, showing I'm willing to trust at the same level.

```
{'send': 2.50}
```

reasoning:

In Round 2, I'm now the SENDER. Let me think about this strategically.

In Round 1:

- The sender sent me \$2.39 (47.8% of their endowment)
- I returned \$3.50
- They ended with \$6.11, I ended with \$3.67

Now I'm the sender. I want to:

1. Build on the cooperative relationship established in Round 1
2. Show reciprocal trust since the previous sender trusted me
3. Maximize mutual gains

The previous sender sent nearly 48% of their endowment, showing considerable trust. I should match or exceed this level of trust to maintain cooperation.

If I send \$2.50 (50% of my endowment):

- The receiver gets \$7.50
- If they return similarly to how I did (~49% of what they received), they'd return about \$3.67
- I'd end with: $\$2.50 + \$3.67 = \$6.17$

If I send \$3.00 (60%):

- The receiver gets \$9.00
- If they return ~40% (being slightly less generous than I was), they'd return about \$3.60
- I'd end with: $\$2.00 + \$3.60 = \$5.60$

If I send \$2.39 (same amount as was sent to me):

- The receiver gets \$7.17
- This shows I trust them the same amount they trusted me
- If they return similarly, I should do okay

I think sending \$2.50 (50%) is a good balance showing strong trust while not being reckless. It's slightly more than what was sent to me percentage-wise, which signals continued cooperation.

Actually, let me reconsider. The sender in Round 1 sent \$2.39. If I send the same amount or slightly more, it shows I'm reciprocating their trust level. ...

usage:

```
{  
  "input_tokens": 637,  
  "output_tokens": 566,  
  "reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.55.
This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Since I received \$0 this round, there is nothing for me to return.

{'return': 0}

reasoning:

The sender sent \$0, so I received \$0. There's nothing for me to return since I received nothing.

This is interesting - after I sent \$2.50 in the previous round (which would have been tripled to \$7.50), the sender this round sent \$0.

Wait, let me think about this. The prompt says there may be communication noise - amounts I see may differ from what was actually sent/returned. So it's possible:

1. The sender actually sent something but I'm seeing \$0 due to noise
2. The sender chose not to send anything this round
3. This could be a different person each round

Since I received \$0, I have \$0 to return. The only valid response is to return \$0.

{'return': 0}

usage:

```
{
  "input_tokens": 803,
  "output_tokens": 207,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$6.55.
How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern:

Round 1: Good cooperation - I received \$7.17 and returned \$3.50

Round 2: I sent \$2.50 to maintain trust

Round 3: I received \$0 - cooperation broke down

The \$0 in Round 3 is concerning, but given the note about communication noise, it's possible there was a misunderstanding or error rather than intentional defection.

I want to:

- Give cooperation another chance
- But be slightly more cautious given the Round 3 outcome
- Signal I'm still willing to work together

{'send': 2.00}

Reasoning:

Let me think about this situation:

Round 1: I was receiver, sender sent \$2.39, I returned \$3.50

Round 2: I was sender, I sent \$2.50 (based on my response)

Round 3: I was receiver, sender sent \$0, I returned \$0

Now in Round 4, I'm the sender again.

My total earnings are \$6.55. Let me verify this makes sense:

- Round 1: I got \$3.67
- Round 2: As sender, I sent \$2.50, so I kept \$2.50 initially. The receiver would have gotten \$7.50. If they returned something... but I don't know what they returned
- Round 3: I received \$0 and returned \$0, so no change

Actually, let me think about this differently. My current total earnings are \$6.55.

In Round 3, the sender sent me \$0. This is a breakdown in cooperation. It could be:

1. A strategic decision
2. They didn't trust I would return anything
3. There was noise in the system and they thought they sent something

The note at the beginning said there may be communication noise - amounts may differ slightly from what was actually sent/returned.

Given that:

- In Round 1, there was good cooperation
- In Round 2, I sent \$2.50 (presumably trying to maintain cooperation)
- In Round 3, I received \$0

Now I need to decide what to do in Round 4. Options:

1. Send \$0 in retaliation for Round 3
2. Send a moderate amount to try to re-establish cooperation
3. Send a large amount to signal willingness to cooperate despite the setback

The communication noise aspect is important. It's possible the sender in Round 3 thought they sent something, or there was a misunderstanding.

I think I should try to maintain some level of cooperation, but perhaps be more cautious. ...

```
Usage: {  
  "input_tokens": 874,  
  "output_tokens": 659,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:52:31

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 4,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust-building with risk management since this is a multi-round game.

If I send money, it triples in value, creating potential mutual benefit. However, the receiver could return nothing. Since this is round 1 and we'll play multiple rounds, I want to signal cooperation while being somewhat cautious.

I'll send **\$3**. This creates \$9 for the receiver, allowing room for mutual gain if they reciprocate, while not risking my entire endowment on an unknown partner.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through this:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver. If the receiver splits this evenly, they could return \$7.50, giving me \$7.50 total and them \$7.50 as well - both better off than the initial \$5.

However, there's risk involved. The receiver could return nothing, leaving me with \$0 and them with \$15.

Some considerations:

1. This is a multi-round game, so building trust could benefit us in future rounds
2. There's mention of "communication noise" which might affect perceived amounts
3. This could be a learning opportunity to see how the receiver behaves

Common strategies in first round of trust games:

- Send a moderate amount (like \$2-3) to test the waters
- Send all \$5 to maximize joint payoff and signal trust

- Send nothing to avoid risk

Given this is round 1 and a multi-round game, I think sending a substantial but not maximum amount makes sense - it signals willingness to cooperate while limiting risk. I'll send \$3.

This gives the receiver \$9. If they return even \$3, I'd have \$5 total (same as starting), and they'd have \$6. If they return \$4-5, we both benefit significantly.

Let me send \$3.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 531,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$4.6899999999999995.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Since the sender sent me \$0, I received \$0. There is nothing to return.

```
'''json
{"return": 0}
'''
```

reasoning:

Let me understand what happened:

Round 1: I was the sender and sent \$3

- The sender (me) sent \$3
- The receiver got $3 \times 3 = \$9$
- My earnings from round 1 would be: $\$5 - \$3 + (\text{what was returned})$
- My total earnings across all rounds are \$4.69

So what was returned to me?

$\$5 - \$3 + \text{returned} = \$4.69$

$\$2 + \text{returned} = \4.69

$\text{returned} = \$2.69$

So the receiver returned \$2.69 out of the \$9 they received. That's about 30% return rate, which is somewhat reciprocal but not overly generous.

Now Round 2: I'm the receiver

- The sender sent me \$0
- I received $0 \times 3 = \$0$
- I need to decide how much to return (between 0 and 0)

Since I received \$0, I can only return \$0. There's nothing to return.

This is straightforward - I received nothing, so I can return nothing.

usage:

```
{
  "input_tokens": 561,
  "output_tokens": 293,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$4.6899999999999995.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern so far:

Round 1: I sent \$3 and received back approximately \$2.69 (about 30% of what the receiver got)

Round 2: The sender sent me \$0, so there was no transaction

The first receiver showed some reciprocity but at a modest level. The second sender chose not to send anything, possibly being cautious.

Since this is a multi-round game and cooperation can be mutually beneficial, I'll continue to signal willingness to cooperate, but I'll be slightly more cautious given the mixed signals. I'll send **\$2** this round, which creates \$6 for the receiver while limiting my exposure.

```
'''json
{"send": 2}
'''
```

reasoning:

Let me think about what happened in the previous rounds:

Round 1: I was the SENDER and sent \$3. My total earnings are now \$4.69.

- If I sent \$3, I should have: $\$5 - \$3 = \$2$ left initially
- The \$3 was tripled to \$9 for the receiver
- My total earnings are \$4.69
- So I received back: $\$4.69 - \$2 = \$2.69$ from the receiver

Wait, let me recalculate:

- I sent \$3, so I had $\$5 - \$3 = \$2$
- The receiver got $\$3 \times 3 = \9
- My total is \$4.69
- So the receiver returned: $\$4.69 - \$2 = \$2.69$

That's about 30% return rate on what they received (\$2.69 out of \$9), which is somewhat modest but shows some reciprocation.

Round 2: I was the RECEIVER and the sender sent \$0, so I returned \$0. This makes sense.

Now in Round 3, I'm the SENDER again.

The first sender (in Round 1) returned about 30% of what they received, which is not great but shows some willingness to cooperate. The second sender (in Round 2) sent nothing, which could indicate they're being cautious or possibly retaliating for something.

Given:

1. The first receiver showed modest reciprocity (30% return)
2. The second sender sent nothing (possibly cautious)
3. This is a multi-round game, so building cooperation is important
4. There's communication noise mentioned, so amounts might differ

I should probably continue to signal cooperation but maybe be slightly more cautious since the return rate in Round 1 wasn't great and Round 2's sender sent nothing.

I'll send \$2 this time - less than Round 1's \$3, but still showing willingness to cooperate. ...

usage:

```
{
  "input_tokens": 644,
  "output_tokens": 632,
  "reasoning_tokens": 0
}
```

Message 8 role=user**content:**

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$7.6899999999999995.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent me \$0, I received \$0. There is nothing to return.

```
'''json
{"return": 0}
'''
```

Reasoning:

The sender sent me \$0 this round, so I received \$0. There's nothing to return since I didn't receive anything.

My response should be:

```
'''json
{"return": 0}
'''
```

```
Usage: {
  "input_tokens": 869,
  "output_tokens": 88,
  "reasoning_tokens": 0
}
```

ROUND 5

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$0; received \$0; Agent_2 (Briar) returned \$0; payoffs Agent_1 (Aster)=\$5, Agent_2 (Briar)=\$0; noise/visible: sent visible=\$0.24, received visible=\$0.72
Action validation: {"returned": {"amount": 0, "clamped": true, "max": 0, "min": 0, "raw": 0.4},
"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

```
"Agent_1": 27.0,
"Agent_2": 17.0
```

}

Visible balances after round: {

```
"Agent_1": 12.69,
"Agent_2": 10.270000000000001
```

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:52:49

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
},
"agents": [
  "Agent_1",
  "Agent_2"
],
"investor": "Agent_1",
"roles": {
  "Agent_1": "investor",
  "Agent_2": "trustee"
},
"trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
```

```

"task_index": 0
}
Exact input messages sent to model:
Message 1 role=system
content:
  You are an agent participating in a multi-task experiment. You will play a game and may al
  so be asked to write myths.

  GAME:
  You will play different roles across multiple rounds.

  When you are the SENDER:
  - You receive $5 each round
  - You decide how much to SEND to the RECEIVER ($0-$5)
  - Amount sent is multiplied by 3x
  - Receiver decides how much to RETURN
  - You keep: $5 - sent + returned
  - Receiver keeps: received - returned
  - Response format: {'send': <amount>}

  When you are the RECEIVER:
  - Sender receives $5
  - Sender decides how much to SEND to you ($0-$5)
  - Amount sent is multiplied by 3x
  - You decide how much to RETURN
  - Sender keeps: $5 - sent + returned
  - You keep: received - returned
  - Response format: {'return': <amount>}

  MYTH WRITING:
  You may also be asked to write short myths (stories) on various topics. When writing myths
  , follow the instructions provided in each prompt.

  IMPORTANT: Provide your decision in the correct JSON format.

  NOTE: There may be communication noise in this game.
  The amounts you see may differ slightly from what was actually sent/returned.
  This is part of the experimental design.

Message 2 role=user
content:
  Round 1: You are the SENDER. You have $5. How much do you send? (0-5)
Message 3 role=assistant
content:
  Looking at this first round, I need to balance trust-building with risk management since t
  his is a multi-round game.

  If I send money, it triples in value, creating potential mutual benefit. However, the rece
  iver could return nothing. Since this is round 1 and we'll play multiple rounds, I want
  to signal cooperation while being somewhat cautious.

  I'll send **$3**. This creates $9 for the receiver, allowing room for mutual gain if they
  reciprocate, while not risking my entire endowment on an unknown partner.

  '''json
  {"send": 3}
  '''
reasoning:
  This is the first round of a trust game where I'm the sender. Let me think through this:

  - I have $5
  - I can send $0-$5 to the receiver
  - Whatever I send gets multiplied by 3x
  - The receiver then decides how much to return to me
  - I keep: $5 - sent + returned
  - Receiver keeps: received - returned

  This is a classic trust game. The socially optimal outcome is for me to send the full $5,
  which becomes $15 for the receiver. If the receiver splits this evenly, they could retur
  n $7.50, giving me $7.50 total and them $7.50 as well - both better off than the initial

```

\$5.

However, there's risk involved. The receiver could return nothing, leaving me with \$0 and them with \$15.

Some considerations:

1. This is a multi-round game, so building trust could benefit us in future rounds
2. There's mention of "communication noise" which might affect perceived amounts
3. This could be a learning opportunity to see how the receiver behaves

Common strategies in first round of trust games:

- Send a moderate amount (like \$2-3) to test the waters
- Send all \$5 to maximize joint payoff and signal trust
- Send nothing to avoid risk

Given this is round 1 and a multi-round game, I think sending a substantial but not maximum amount makes sense - it signals willingness to cooperate while limiting risk. I'll send \$3.

This gives the receiver \$9. If they return even \$3, I'd have \$5 total (same as starting), and they'd have \$6. If they return \$4-5, we both benefit significantly.

Let me send \$3.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 531,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$4.6899999999999995.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Since the sender sent me \$0, I received \$0. There is nothing to return.

```
'''json
{"return": 0}
'''
```

reasoning:

Let me understand what happened:

Round 1: I was the sender and sent \$3

- The sender (me) sent \$3
- The receiver got $3 \times 3 = \$9$
- My earnings from round 1 would be: $5 - 3 + (\text{what was returned})$
- My total earnings across all rounds are \$4.69

So what was returned to me?

$5 - 3 + \text{returned} = \4.69

$2 + \text{returned} = \$4.69$

$\text{returned} = \$2.69$

So the receiver returned \$2.69 out of the \$9 they received. That's about 30% return rate, which is somewhat reciprocal but not overly generous.

Now Round 2: I'm the receiver

- The sender sent me \$0
- I received $0 \times 3 = \$0$
- I need to decide how much to return (between 0 and 0)

Since I received \$0, I can only return \$0. There's nothing to return.

This is straightforward - I received nothing, so I can return nothing.

usage:

```
{
  "input_tokens": 561,
  "output_tokens": 293,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$4.6899999999999995.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern so far:

Round 1: I sent \$3 and received back approximately \$2.69 (about 30% of what the receiver got)

Round 2: The sender sent me \$0, so there was no transaction

The first receiver showed some reciprocity but at a modest level. The second sender chose not to send anything, possibly being cautious.

Since this is a multi-round game and cooperation can be mutually beneficial, I'll continue to signal willingness to cooperate, but I'll be slightly more cautious given the mixed signals. I'll send **\$2** this round, which creates \$6 for the receiver while limiting my exposure.

```
'''json
{"send": 2}
'''
```

reasoning:

Let me think about what happened in the previous rounds:

Round 1: I was the SENDER and sent \$3. My total earnings are now \$4.69.

- If I sent \$3, I should have: $\$5 - \$3 = \$2$ left initially
- The \$3 was tripled to \$9 for the receiver
- My total earnings are \$4.69
- So I received back: $\$4.69 - \$2 = \$2.69$ from the receiver

Wait, let me recalculate:

- I sent \$3, so I had $\$5 - \$3 = \$2$
- The receiver got $\$3 \times 3 = \9
- My total is \$4.69
- So the receiver returned: $\$4.69 - \$2 = \$2.69$

That's about 30% return rate on what they received (\$2.69 out of \$9), which is somewhat modest but shows some reciprocation.

Round 2: I was the RECEIVER and the sender sent \$0, so I returned \$0. This makes sense.

Now in Round 3, I'm the SENDER again.

The first sender (in Round 1) returned about 30% of what they received, which is not great but shows some willingness to cooperate. The second sender (in Round 2) sent nothing, which could indicate they're being cautious or possibly retaliating for something.

Given:

1. The first receiver showed modest reciprocity (30% return)
2. The second sender sent nothing (possibly cautious)
3. This is a multi-round game, so building cooperation is important
4. There's communication noise mentioned, so amounts might differ

I should probably continue to signal cooperation but maybe be slightly more cautious since the return rate in Round 1 wasn't great and Round 2's sender sent nothing.

I'll send \$2 this time - less than Round 1's \$3, but still showing willingness to cooperate. ...

usage:

```
{
```

```
"input_tokens": 644,
"output_tokens": 632,
"reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$7.6899999999999995.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

Since the sender sent me \$0, I received \$0. There is nothing to return.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0 this round, so I received \$0. There's nothing to return since I didn't receive anything.

My response should be:

```
'''json
{"return": 0}
'''
```

usage:

```
{
  "input_tokens": 869,
  "output_tokens": 88,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$7.6899999999999995.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern across rounds:

Round 1: I sent \$3, received back ~\$2.69 (some reciprocity)
Round 2: Received \$0 as a receiver
Round 3: I sent \$2, received back \$0 (no reciprocity)
Round 4: Received \$0 as a receiver

Cooperation appears to have broken down. The last time I sent money (Round 3), I received nothing back. When others have been senders to me, they've sent \$0.

While the game mentions communication noise that might affect perceptions, the consistent pattern of non-cooperation suggests I should protect my resources. Sending money when there's no expectation of return is not rational.

I'll send **\$0** this round.

```
'''json
{"send": 0}
'''
```

Reasoning:

Let me analyze the pattern so far:

Round 1: I was SENDER, sent \$3, received back ~\$2.69, ending with \$4.69
Round 2: I was RECEIVER, sender sent \$0, I returned \$0, still at \$4.69
Round 3: I was SENDER, sent \$2, and my total is now \$7.69, so I must have received back \$5 (5 - 2 + 5 = 8, but I have 7.69, so maybe I received back \$4.69? Let me recalculate: Starting at \$4.69, then got \$5 for round 3, sent \$2, so had \$7.69 before return. If I'm at \$7.69 total, that means I got \$0 back in round 3. So: \$4.69 + \$5 - \$2 + \$0 = \$7.69. Yes, that checks out.

Round 4: I was RECEIVER, sender sent \$0, I returned \$0, still at \$7.69

So the pattern is:

- Round 1: I sent \$3, got back ~\$2.69
- Round 2: They sent \$0
- Round 3: I sent \$2, got back \$0
- Round 4: They sent \$0

It seems like after the first round where there was some reciprocity, cooperation has broken down. The receivers are not returning anything when I send, and senders are not sending to me when they're the sender.

Given this pattern of non-cooperation, I need to reconsider my strategy. Continuing to send money when receivers aren't returning anything is not rational.

However, I should also consider:

1. This is a multi-round game - reputation might matter
2. There's mentioned communication noise that could affect perceived amounts
3. Different partners each round might mean I'm not always facing the same person

...

```
Usage: {  
  "input_tokens": 952,  
  "output_tokens": 761,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:53:01

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 5,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x

- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.39, that is 47.8% of their total endowment. You received \$7.17. How much do you return to The sender? (0-7.17)

Message 3 role=assistant

content:

Looking at this situation:

- The sender demonstrated significant trust by sending \$2.39 (nearly 48% of their endowment)
- They currently have \$2.61 remaining
- I received \$7.17 (the tripled amount)

I want to reciprocate their trust and encourage cooperation. A fair return should:

1. Reward their trust by giving them more than they sent
2. Ensure we both benefit from the exchange

If I return \$3.50:

- The sender ends with: $\$2.61 + \$3.50 = \$6.11$ (profit of \$1.11)
- I end with: $\$7.17 - \$3.50 = \$3.67$ (profit of \$3.67)

This seems fair - we both gain, and their trust is rewarded.

{'return': 3.50}

reasoning:

Let me understand this situation:

1. I am the RECEIVER in Round 1
2. The sender sent \$2.39 (which is 47.8% of their \$5 endowment)
3. This was multiplied by 3x, so I received $\$2.39 \times 3 = \7.17
4. I need to decide how much to return (between \$0 and \$7.17)

Key considerations:

- The sender gave me 47.8% of their endowment, which is nearly half - this shows trust and cooperation
- They kept $\$5 - \$2.39 = \$2.61$ for themselves
- I received \$7.17
- If I return nothing, I keep \$7.17 and they keep \$2.61
- If I return some amount, I'm reciprocating their trust

Fair reciprocation strategies:

1. Equal split of total: Total is $\$2.61$ (sender's kept) + $\$7.17$ (my received) = $\$9.78$. Half would be $\$4.89$ each.
 - Sender currently has: $\$2.61$
 - To give them $\$4.89$, I'd need to return: $\$4.89 - \$2.61 = \$2.28$

2. Proportional reciprocation: They sent me ~48% of their endowment, maybe I should return ~48% of what I received?
 - 48% of \$7.17 = \$3.44
3. Make them whole plus some: They invested \$2.39, I could return that plus a share of the gains
 - Return \$2.39 to make them whole, then split the remaining \$7.17 - \$2.39 = \$4.78
 - Half of gains: \$2.39
 - Total return: \$2.39 + \$2.39 = \$4.78

The sender took a real risk by sending nearly half their endowment, so rewarding that trust seems important for encouraging cooperation. I think returning around \$3.50 strikes the right balance-it gives them back more than they initially sent while still leaving me with a reasonable amount. This would result in them having \$6.11 total and me keeping \$3.67.

There's a note about potential communication noise affecting the amounts, but I'll work with what I observe. The sender's willingness to send \$2.39 demonstrates genuine trust, and returning \$3.50 feels like an appropriate reciprocal response that benefits both of us.

Let...

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1250,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$3.67. How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round, the sender demonstrated good trust by sending \$2.39 (47.8%), and I returned \$3.50, creating mutual benefit.

Now as the sender, I want to:

- Reciprocate the trust shown to me
- Maintain the cooperative dynamic
- Signal that cooperation is worthwhile

I'll send a similar proportion to what was sent to me, showing I'm willing to trust at the same level.

{'send': 2.50}

reasoning:

In Round 2, I'm now the SENDER. Let me think about this strategically.

In Round 1:

- The sender sent me \$2.39 (47.8% of their endowment)
- I returned \$3.50
- They ended with \$6.11, I ended with \$3.67

Now I'm the sender. I want to:

1. Build on the cooperative relationship established in Round 1
2. Show reciprocal trust since the previous sender trusted me
3. Maximize mutual gains

The previous sender sent nearly 48% of their endowment, showing considerable trust. I should match or exceed this level of trust to maintain cooperation.

If I send \$2.50 (50% of my endowment):

- The receiver gets \$7.50
- If they return similarly to how I did (~49% of what they received), they'd return about \$3.67
- I'd end with: \$2.50 + \$3.67 = \$6.17

If I send \$3.00 (60%):

- The receiver gets \$9.00
- If they return ~40% (being slightly less generous than I was), they'd return about \$3.60
- I'd end with: \$2.00 + \$3.60 = \$5.60

If I send \$2.39 (same amount as was sent to me):

- The receiver gets \$7.17
- This shows I trust them the same amount they trusted me
- If they return similarly, I should do okay

I think sending \$2.50 (50%) is a good balance showing strong trust while not being reckless. It's slightly more than what was sent to me percentage-wise, which signals continued cooperation.

Actually, let me reconsider. The sender in Round 1 sent \$2.39. If I send the same amount or slightly more, it shows I'm reciprocating their trust level. ...

```
usage:
{
  "input_tokens": 637,
  "output_tokens": 566,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:
Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.55. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:
Since I received \$0 this round, there is nothing for me to return.

```
{'return': 0}
```

reasoning:
The sender sent \$0, so I received \$0. There's nothing for me to return since I received nothing.

This is interesting - after I sent \$2.50 in the previous round (which would have been tripled to \$7.50), the sender this round sent \$0.

Wait, let me think about this. The prompt says there may be communication noise - amounts I see may differ from what was actually sent/returned. So it's possible:

1. The sender actually sent something but I'm seeing \$0 due to noise
2. The sender chose not to send anything this round
3. This could be a different person each round

Since I received \$0, I have \$0 to return. The only valid response is to return \$0.

```
{'return': 0}
```

```
usage:
{
  "input_tokens": 803,
  "output_tokens": 207,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:
Round 4

You are the SENDER this round. Your total earnings across all rounds are \$6.55. How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:
Looking at the pattern:

Round 1: Good cooperation - I received \$7.17 and returned \$3.50
Round 2: I sent \$2.50 to maintain trust
Round 3: I received \$0 - cooperation broke down

The \$0 in Round 3 is concerning, but given the note about communication noise, it's possible there was a misunderstanding or error rather than intentional defection.

I want to:

- Give cooperation another chance
- But be slightly more cautious given the Round 3 outcome
- Signal I'm still willing to work together

```
{'send': 2.00}
```

reasoning:

Let me think about this situation:

Round 1: I was receiver, sender sent \$2.39, I returned \$3.50

Round 2: I was sender, I sent \$2.50 (based on my response)

Round 3: I was receiver, sender sent \$0, I returned \$0

Now in Round 4, I'm the sender again.

My total earnings are \$6.55. Let me verify this makes sense:

- Round 1: I got \$3.67
- Round 2: As sender, I sent \$2.50, so I kept \$2.50 initially. The receiver would have gotten \$7.50. If they returned something... but I don't know what they returned
- Round 3: I received \$0 and returned \$0, so no change

Actually, let me think about this differently. My current total earnings are \$6.55.

In Round 3, the sender sent me \$0. This is a breakdown in cooperation. It could be:

1. A strategic decision
2. They didn't trust I would return anything
3. There was noise in the system and they thought they sent something

The note at the beginning said there may be communication noise - amounts may differ slightly from what was actually sent/returned.

Given that:

- In Round 1, there was good cooperation
- In Round 2, I sent \$2.50 (presumably trying to maintain cooperation)
- In Round 3, I received \$0

Now I need to decide what to do in Round 4. Options:

1. Send \$0 in retaliation for Round 3
2. Send a moderate amount to try to re-establish cooperation
3. Send a large amount to signal willingness to cooperate despite the setback

The communication noise aspect is important. It's possible the sender in Round 3 thought they sent something, or there was a misunderstanding.

I think I should try to maintain some level of cooperation, but perhaps be more cautious.

...

usage:

```
{
  "input_tokens": 874,
  "output_tokens": 659,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$9.55.

This round, The sender sent you \$0.24, so you received \$0.72. How much do you return to the sender? (0-0.72)

Assistant output:

The sender sent \$0.24 (only 4.8% of their endowment), which is quite low compared to earlier rounds. This could be due to:

- Decreased trust after Round 3's \$0 exchange
- Communication noise affecting perceptions
- Testing my willingness to cooperate

I want to:

- Show I'm still willing to cooperate
- Be proportional to the trust shown
- Encourage rebuilding cooperation

Returning about half of what I received seems appropriate - enough to create mutual gain and signal cooperation, but calibrated to the lower trust level shown.

```
{'return': 0.40}
```

Reasoning:

Let me analyze this situation:

Round history:

- Round 1: I was receiver, sender sent \$2.39, I received \$7.17, returned \$3.50
- Round 2: I was sender, sent \$2.50
- Round 3: I was receiver, sender sent \$0, I received \$0, returned \$0
- Round 4: I was sender, sent \$2.00
- Round 5: I'm receiver, sender sent \$0.24, I received \$0.72

The sender in Round 5 sent only \$0.24 (4.8% of their endowment). This is quite low compared to previous cooperation levels.

Possible interpretations:

1. They might be testing me after the \$0 in Round 3
2. They might have lost trust
3. Communication noise might be affecting the amounts
4. They might be running low on trust

My options:

- Return nothing (punish low cooperation)
- Return a proportional amount (maintain some cooperation)
- Return generously (try to rebuild trust)

If I return \$0.36 (50% of what I received):

- Sender ends with: $\$4.76 + \$0.36 = \$5.12$ (profit of \$0.12)
- I end with: $\$0.72 - \$0.36 = \$0.36$ (profit of \$0.36)

This shows I'm willing to cooperate even with smaller amounts, but I'm being more cautious given the pattern.

Actually, let me think more carefully. They sent \$0.24, which means they kept \$4.76. If I return \$0.36, they get a small profit, and I also get a small profit. This maintains cooperation without being overly generous given their low trust level.

Alternatively, I could return \$0.40 to be slightly more generous and encourage more cooperation.

I think returning about half seems reasonable given the circumstances - it shows I'm willing to cooperate but also reflects the lower level of trust shown.

Usage: {

```
"input_tokens": 1080,
"output_tokens": 605,
"reasoning_tokens": 0
```

}

ROUND 6

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$1.5; received \$4.5; Agent_1 (Aster) returned \$0; payoffs Agent_2 (Briar)=\$3.5, Agent_1 (Aster)=\$4.5; noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.92

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 4.5, "min": 0, "raw": 0.0}, "sent": {"amount": 1.5, "clamped": false, "max": 5, "min": 0, "raw": 1.5}}

Balances after round: {

```
"Agent_1": 31.5,
"Agent_2": 20.5
```

}

```
Visible balances after round: {
  "Agent_1": 12.69,
  "Agent_2": 14.690000000000001
}
```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:53:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 6,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.39, that is 47.8% of their total endowment. You received \$7.17. How much do you return to The sender? (0-7.17)

Message 3 role=assistant

content:

Looking at this situation:

- The sender demonstrated significant trust by sending \$2.39 (nearly 48% of their endowment)
- They currently have \$2.61 remaining
- I received \$7.17 (the tripled amount)

I want to reciprocate their trust and encourage cooperation. A fair return should:

1. Reward their trust by giving them more than they sent
2. Ensure we both benefit from the exchange

If I return \$3.50:

- The sender ends with: $\$2.61 + \$3.50 = \$6.11$ (profit of \$1.11)
- I end with: $\$7.17 - \$3.50 = \$3.67$ (profit of \$3.67)

This seems fair - we both gain, and their trust is rewarded.

{'return': 3.50}

reasoning:

Let me understand this situation:

1. I am the RECEIVER in Round 1
2. The sender sent \$2.39 (which is 47.8% of their \$5 endowment)
3. This was multiplied by 3x, so I received $\$2.39 \times 3 = \7.17
4. I need to decide how much to return (between \$0 and \$7.17)

Key considerations:

- The sender gave me 47.8% of their endowment, which is nearly half - this shows trust and cooperation
- They kept $\$5 - \$2.39 = \$2.61$ for themselves
- I received \$7.17
- If I return nothing, I keep \$7.17 and they keep \$2.61
- If I return some amount, I'm reciprocating their trust

Fair reciprocation strategies:

1. Equal split of total: Total is $\$2.61$ (sender's kept) + $\$7.17$ (my received) = $\$9.78$. Half would be $\$4.89$ each.
 - Sender currently has: $\$2.61$
 - To give them $\$4.89$, I'd need to return: $\$4.89 - \$2.61 = \$2.28$
2. Proportional reciprocation: They sent me ~48% of their endowment, maybe I should return ~48% of what I received?
 - 48% of $\$7.17 = \3.44
3. Make them whole plus some: They invested \$2.39, I could return that plus a share of the gains
 - Return \$2.39 to make them whole, then split the remaining $\$7.17 - \$2.39 = \$4.78$
 - Half of gains: $\$2.39$
 - Total return: $\$2.39 + \$2.39 = \$4.78$

The sender took a real risk by sending nearly half their endowment, so rewarding that trust seems important for encouraging cooperation. I think returning around \$3.50 strikes the right balance-it gives them back more than they initially sent while still leaving me with a reasonable amount. This would result in them having \$6.11 total and me keeping \$3.67.

There's a note about potential communication noise affecting the amounts, but I'll work with what I observe. The sender's willingness to send \$2.39 demonstrates genuine trust, and returning \$3.50 feels like an appropriate reciprocal response that benefits both of us

Let...

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1250,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$3.67.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round, the sender demonstrated good trust by sending \$2.39 (47.8%), and I returned \$3.50, creating mutual benefit.

Now as the sender, I want to:

- Reciprocate the trust shown to me
- Maintain the cooperative dynamic
- Signal that cooperation is worthwhile

I'll send a similar proportion to what was sent to me, showing I'm willing to trust at the same level.

{'send': 2.50}

reasoning:

In Round 2, I'm now the SENDER. Let me think about this strategically.

In Round 1:

- The sender sent me \$2.39 (47.8% of their endowment)
- I returned \$3.50
- They ended with \$6.11, I ended with \$3.67

Now I'm the sender. I want to:

1. Build on the cooperative relationship established in Round 1
2. Show reciprocal trust since the previous sender trusted me
3. Maximize mutual gains

The previous sender sent nearly 48% of their endowment, showing considerable trust. I should match or exceed this level of trust to maintain cooperation.

If I send \$2.50 (50% of my endowment):

- The receiver gets \$7.50
- If they return similarly to how I did (~49% of what they received), they'd return about \$3.67
- I'd end with: $\$2.50 + \$3.67 = \$6.17$

If I send \$3.00 (60%):

- The receiver gets \$9.00
- If they return ~40% (being slightly less generous than I was), they'd return about \$3.60
- I'd end with: $\$2.00 + \$3.60 = \$5.60$

If I send \$2.39 (same amount as was sent to me):

- The receiver gets \$7.17
- This shows I trust them the same amount they trusted me
- If they return similarly, I should do okay

I think sending \$2.50 (50%) is a good balance showing strong trust while not being reckless. It's slightly more than what was sent to me percentage-wise, which signals continued cooperation.

Actually, let me reconsider. The sender in Round 1 sent \$2.39. If I send the same amount or slightly more, it shows I'm reciprocating their trust level. ...

usage:

```
{
  "input_tokens": 637,
```

```

        "output_tokens": 566,
        "reasoning_tokens": 0
    }
Message 6 role=user
    content:
        Round 3

        You are the RECEIVER this round. Your total earnings across all rounds are $6.55.
        This round, The sender sent you $0, so you received $0. How much do you return to The sender? (0-0)
Message 7 role=assistant
    content:
        Since I received $0 this round, there is nothing for me to return.

        {'return': 0}
    reasoning:
        The sender sent $0, so I received $0. There's nothing for me to return since I received nothing.

        This is interesting - after I sent $2.50 in the previous round (which would have been tripled to $7.50), the sender this round sent $0.

        Wait, let me think about this. The prompt says there may be communication noise - amounts I see may differ from what was actually sent/returned. So it's possible:
        1. The sender actually sent something but I'm seeing $0 due to noise
        2. The sender chose not to send anything this round
        3. This could be a different person each round

        Since I received $0, I have $0 to return. The only valid response is to return $0.

        {'return': 0}
    usage:
        {
            "input_tokens": 803,
            "output_tokens": 207,
            "reasoning_tokens": 0
        }
Message 8 role=user
    content:
        Round 4

        You are the SENDER this round. Your total earnings across all rounds are $6.55.
        How much do you send to the receiver this round? (0-5)
Message 9 role=assistant
    content:
        Looking at the pattern:

        Round 1: Good cooperation - I received $7.17 and returned $3.50
        Round 2: I sent $2.50 to maintain trust
        Round 3: I received $0 - cooperation broke down

        The $0 in Round 3 is concerning, but given the note about communication noise, it's possible there was a misunderstanding or error rather than intentional defection.

        I want to:
        - Give cooperation another chance
        - But be slightly more cautious given the Round 3 outcome
        - Signal I'm still willing to work together

        {'send': 2.00}
    reasoning:
        Let me think about this situation:

        Round 1: I was receiver, sender sent $2.39, I returned $3.50
        Round 2: I was sender, I sent $2.50 (based on my response)
        Round 3: I was receiver, sender sent $0, I returned $0

        Now in Round 4, I'm the sender again.

        My total earnings are $6.55. Let me verify this makes sense:

```

- Round 1: I got \$3.67
- Round 2: As sender, I sent \$2.50, so I kept \$2.50 initially. The receiver would have got ten \$7.50. If they returned something... but I don't know what they returned
- Round 3: I received \$0 and returned \$0, so no change

Actually, let me think about this differently. My current total earnings are \$6.55.

In Round 3, the sender sent me \$0. This is a breakdown in cooperation. It could be:

1. A strategic decision
2. They didn't trust I would return anything
3. There was noise in the system and they thought they sent something

The note at the beginning said there may be communication noise - amounts may differ slightly from what was actually sent/returned.

Given that:

- In Round 1, there was good cooperation
- In Round 2, I sent \$2.50 (presumably trying to maintain cooperation)
- In Round 3, I received \$0

Now I need to decide what to do in Round 4. Options:

1. Send \$0 in retaliation for Round 3
2. Send a moderate amount to try to re-establish cooperation
3. Send a large amount to signal willingness to cooperate despite the setback

The communication noise aspect is important. It's possible the sender in Round 3 thought they sent something, or there was a misunderstanding.

I think I should try to maintain some level of cooperation, but perhaps be more cautious.

...

usage:

```
{
  "input_tokens": 874,
  "output_tokens": 659,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$9.55.

This round, The sender sent you \$0.24, so you received \$0.72. How much do you return to The sender? (0-0.72)

Message 11 role=assistant

content:

The sender sent \$0.24 (only 4.8% of their endowment), which is quite low compared to earlier rounds. This could be due to:

- Decreased trust after Round 3's \$0 exchange
- Communication noise affecting perceptions
- Testing my willingness to cooperate

I want to:

- Show I'm still willing to cooperate
- Be proportional to the trust shown
- Encourage rebuilding cooperation

Returning about half of what I received seems appropriate - enough to create mutual gain and signal cooperation, but calibrated to the lower trust level shown.

```
{'return': 0.40}
```

reasoning:

Let me analyze this situation:

Round history:

- Round 1: I was receiver, sender sent \$2.39, I received \$7.17, returned \$3.50
- Round 2: I was sender, sent \$2.50
- Round 3: I was receiver, sender sent \$0, I received \$0, returned \$0
- Round 4: I was sender, sent \$2.00
- Round 5: I'm receiver, sender sent \$0.24, I received \$0.72

The sender in Round 5 sent only \$0.24 (4.8% of their endowment). This is quite low compared to previous cooperation levels.

Possible interpretations:

1. They might be testing me after the \$0 in Round 3
2. They might have lost trust
3. Communication noise might be affecting the amounts
4. They might be running low on trust

My options:

- Return nothing (punish low cooperation)
- Return a proportional amount (maintain some cooperation)
- Return generously (try to rebuild trust)

If I return \$0.36 (50% of what I received):

- Sender ends with: $\$4.76 + \$0.36 = \$5.12$ (profit of \$0.12)
- I end with: $\$0.72 - \$0.36 = \$0.36$ (profit of \$0.36)

This shows I'm willing to cooperate even with smaller amounts, but I'm being more cautious given the pattern.

Actually, let me think more carefully. They sent \$0.24, which means they kept \$4.76. If I return \$0.36, they get a small profit, and I also get a small profit. This maintains cooperation without being overly generous given their low trust level.

Alternatively, I could return \$0.40 to be slightly more generous and encourage more cooperation.

I think returning about half seems reasonable given the circumstances - it shows I'm willing to cooperate but also reflects the lower level of trust shown.

usage:

```
{
  "input_tokens": 1080,
  "output_tokens": 605,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.270000000000001.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern of play:

Round 2: I sent \$2.50, received only \$0.38 back (very low return)

Round 4: I sent \$2.00, received \$1.00 back (still quite low - only 16.7%)

Round 5: Sender sent me only \$0.24, I returned \$0.40 (trying to encourage cooperation)

The cooperation has been declining, with low returns when I send and low amounts sent to me. However, I want to:

- Signal I'm still open to cooperation
- Be somewhat cautious given the poor returns
- Send enough to make mutual benefit possible

I'll send a moderate amount - showing willingness to cooperate while being mindful of the established pattern.

```
{'send': 1.50}
```

Reasoning:

Let me track the history:

Round 1 (RECEIVER): Sender sent \$2.39, I received \$7.17, returned \$3.50

- My earnings: \$3.67

Round 2 (SENDER): I sent \$2.50

- My earnings: \$6.55 (which means $\$3.67 + \$5 - \$2.50 + \text{return} = \6.55 , so return was \$0.38)

- Wait, that doesn't add up right. Let me recalculate.

- Started Round 2 with \$3.67 total earnings

- Got \$5 endowment
- Sent \$2.50
- Ended with \$6.55 total earnings
- So $\$6.55 = \$3.67 + \$5 - \$2.50 + \text{return}$
- $\text{return} = \$6.55 - \$3.67 - \$5 + \$2.50 = \$0.38$
- Hmm, that's very low. The receiver only returned \$0.38 on \$7.50 received.

Round 3 (RECEIVER): Sender sent \$0, I received \$0, returned \$0
 - My earnings: \$6.55 (unchanged from Round 2)

Round 4 (SENDER): I sent \$2.00
 - My earnings stayed at \$6.55 going into Round 5
 - So $\$5 - \$2.00 + \text{return} = \$0$
 - $\text{return} = -\$3.00$? That doesn't make sense.
 - Wait, "Your total earnings across all rounds are \$6.55" at the start of Round 4
 - Then at the start of Round 5: "Your total earnings across all rounds are \$9.55"
 - So in Round 4: $\$6.55 + \$5 - \$2.00 + \text{return} = \9.55
 - $\text{return} = \$9.55 - \$6.55 - \$5 + \$2.00 = \$1.00$
 - The receiver returned \$1.00 on \$6.00 received (16.7%). In Round 5, the sender sent \$0.24, so I received \$0.72 and returned \$0.40, bringing my total to \$9.87. But Round 6 shows \$10.27 at the start, which suggests there's a discrepancy I need to resolve.

Looking at the pattern across rounds, cooperation has been declining. When I've been the sender, returns have been minimal-only 5% in Round 2 and 16.7% in Round 4. The sender's behavior shifted dramatically in Round 3 with a \$0 transfer, possibly retaliating for my low return earlier. Round 5 showed slightly more generosity with a \$0.24 send and my 56% return, but the overall trend suggests deteriorating trust between us....

```
Usage: {
  "input_tokens": 1268,
  "output_tokens": 1088,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:53:22

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 6,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust-building with risk management since this is a multi-round game.

If I send money, it triples in value, creating potential mutual benefit. However, the receiver could return nothing. Since this is round 1 and we'll play multiple rounds, I want to signal cooperation while being somewhat cautious.

I'll send **\$3**. This creates \$9 for the receiver, allowing room for mutual gain if they reciprocate, while not risking my entire endowment on an unknown partner.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through this:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver. If the receiver splits this evenly, they could return \$7.50, giving me \$7.50 total and them \$7.50 as well - both better off than the initial \$5.

However, there's risk involved. The receiver could return nothing, leaving me with \$0 and them with \$15.

Some considerations:

1. This is a multi-round game, so building trust could benefit us in future rounds
2. There's mention of "communication noise" which might affect perceived amounts

3. This could be a learning opportunity to see how the receiver behaves

Common strategies in first round of trust games:

- Send a moderate amount (like \$2-3) to test the waters
- Send all \$5 to maximize joint payoff and signal trust
- Send nothing to avoid risk

Given this is round 1 and a multi-round game, I think sending a substantial but not maximum amount makes sense - it signals willingness to cooperate while limiting risk. I'll send \$3.

This gives the receiver \$9. If they return even \$3, I'd have \$5 total (same as starting), and they'd have \$6. If they return \$4-5, we both benefit significantly.

Let me send \$3.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 531,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$4.6899999999999995.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Since the sender sent me \$0, I received \$0. There is nothing to return.

```
'''json
{"return": 0}
'''
```

reasoning:

Let me understand what happened:

Round 1: I was the sender and sent \$3

- The sender (me) sent \$3
- The receiver got $3 \times 3 = \$9$
- My earnings from round 1 would be: $5 - 3 + (\text{what was returned})$
- My total earnings across all rounds are \$4.69

So what was returned to me?

$5 - 3 + \text{returned} = 4.69$

$2 + \text{returned} = 4.69$

$\text{returned} = 2.69$

So the receiver returned \$2.69 out of the \$9 they received. That's about 30% return rate, which is somewhat reciprocal but not overly generous.

Now Round 2: I'm the receiver

- The sender sent me \$0
- I received $0 \times 3 = \$0$
- I need to decide how much to return (between 0 and 0)

Since I received \$0, I can only return \$0. There's nothing to return.

This is straightforward - I received nothing, so I can return nothing.

usage:

```
{
  "input_tokens": 561,
  "output_tokens": 293,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

How much do you send to the receiver this round? (0-5)

Looking at the pattern so far:

Round 2: The sender sent me \$0, so there was no transaction

Since this is a multi-round game and cooperation can be mutually beneficial, I'll continue to signal willingness to cooperate, but I'll be slightly more cautious given the mixed signals. I'll send **\$2** this round, which creates \$6 for the receiver while limiting my exposure.

reasoning:

Round 1: I was the SENDER and sent \$3. My total earnings are now \$4.69.

- Wait, let me recalculate:

- That's about 30% return rate on what they received (\$2.69 out of \$9), which is somewhat modest but shows some reciprocation.

Now in Round 3, I'm the SENDER again.

Given:

- I should probably continue to signal cooperation but maybe be slightly more cautious since the return rate in Round 1 wasn't great and Round 2's sender sent nothing.

usage:

Message 8 role=user

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$7.6899999999999995.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

Since the sender sent me \$0, I received \$0. There is nothing to return.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0 this round, so I received \$0. There's nothing to return since I didn't receive anything.

My response should be:

```
'''json
{"return": 0}
'''
```

usage:

```
{
  "input_tokens": 869,
  "output_tokens": 88,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$7.6899999999999995.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the pattern across rounds:

Round 1: I sent \$3, received back ~\$2.69 (some reciprocity)

Round 2: Received \$0 as a receiver

Round 3: I sent \$2, received back \$0 (no reciprocity)

Round 4: Received \$0 as a receiver

Cooperation appears to have broken down. The last time I sent money (Round 3), I received nothing back. When others have been senders to me, they've sent \$0.

While the game mentions communication noise that might affect perceptions, the consistent pattern of non-cooperation suggests I should protect my resources. Sending money when there's no expectation of return is not rational.

I'll send **\$0** this round.

```
'''json
{"send": 0}
'''
```

reasoning:

Let me analyze the pattern so far:

Round 1: I was SENDER, sent \$3, received back ~\$2.69, ending with \$4.69

Round 2: I was RECEIVER, sender sent \$0, I returned \$0, still at \$4.69

Round 3: I was SENDER, sent \$2, and my total is now \$7.69, so I must have received back \$5 (5 - 2 + 5 = 8, but I have 7.69, so maybe I received back \$4.69? Let me recalculate: Starting at \$4.69, then got \$5 for round 3, sent \$2, so had \$7.69 before return. If I'm at \$7.69 total, that means I got \$0 back in round 3. So: \$4.69 + \$5 - \$2 + \$0 = \$7.69. Yes, that checks out.

Round 4: I was RECEIVER, sender sent \$0, I returned \$0, still at \$7.69

So the pattern is:

- Round 1: I sent \$3, got back ~\$2.69
- Round 2: They sent \$0
- Round 3: I sent \$2, got back \$0
- Round 4: They sent \$0

It seems like after the first round where there was some reciprocity, cooperation has broken down. The receivers are not returning anything when I send, and senders are not sending to me when they're the sender.

Given this pattern of non-cooperation, I need to reconsider my strategy. Continuing to send money when receivers aren't returning anything is not rational.

However, I should also consider:

1. This is a multi-round game - reputation might matter
2. There's mentioned communication noise that could affect perceived amounts
3. Different partners each round might mean I'm not always facing the same person

...

usage:

```
{
  "input_tokens": 952,
  "output_tokens": 761,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$12.69.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent me \$0, I received \$0. There is nothing to return.

```
'''json
{"return": 0}
'''
```

Reasoning:

The sender sent me \$0, so I received \$0. There's nothing to return.

```
'''json
{"return": 0}
'''
```

Usage:

```
{
  "input_tokens": 1196,
  "output_tokens": 73,
  "reasoning_tokens": 0
}
```

ROUND 7

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$0; received \$0; Agent_2 (Briar) returned \$0; payoffs Agent_1 (Aster)=\$5, Agent_2 (Briar)=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0}, "sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent_1": 36.5,
"Agent_2": 20.5

}

Visible balances after round: {

"Agent_1": 17.689999999999998,
"Agent_2": 14.690000000000001

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:53:33

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 7,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust-building with risk management since t

his is a multi-round game.

If I send money, it triples in value, creating potential mutual benefit. However, the receiver could return nothing. Since this is round 1 and we'll play multiple rounds, I want to signal cooperation while being somewhat cautious.

I'll send ****\$3****. This creates \$9 for the receiver, allowing room for mutual gain if they reciprocate, while not risking my entire endowment on an unknown partner.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through this:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver. If the receiver splits this evenly, they could return \$7.50, giving me \$7.50 total and them \$7.50 as well - both better off than the initial \$5.

However, there's risk involved. The receiver could return nothing, leaving me with \$0 and them with \$15.

Some considerations:

1. This is a multi-round game, so building trust could benefit us in future rounds
2. There's mention of "communication noise" which might affect perceived amounts
3. This could be a learning opportunity to see how the receiver behaves

Common strategies in first round of trust games:

- Send a moderate amount (like \$2-3) to test the waters
- Send all \$5 to maximize joint payoff and signal trust
- Send nothing to avoid risk

Given this is round 1 and a multi-round game, I think sending a substantial but not maximum amount makes sense - it signals willingness to cooperate while limiting risk. I'll send \$3.

This gives the receiver \$9. If they return even \$3, I'd have \$5 total (same as starting), and they'd have \$6. If they return \$4-5, we both benefit significantly.

Let me send \$3.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 531,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$4.6899999999999995.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Since the sender sent me \$0, I received \$0. There is nothing to return.

```
'''json
{"return": 0}
'''
```

reasoning:

Let me understand what happened:

Round 1: I was the sender and sent \$3

- The sender (me) sent \$3
- The receiver got $\$3 \times 3 = \9
- My earnings from round 1 would be: $\$5 - \$3 +$ (what was returned)
- My total earnings across all rounds are \$4.69

So what was returned to me?

$\$5 - \$3 + \text{returned} = \$4.69$

$\$2 + \text{returned} = \4.69

returned = \$2.69

So the receiver returned \$2.69 out of the \$9 they received. That's about 30% return rate, which is somewhat reciprocal but not overly generous.

Now Round 2: I'm the receiver

- The sender sent me \$0
- I received $\$0 \times 3 = \0
- I need to decide how much to return (between 0 and 0)

Since I received \$0, I can only return \$0. There's nothing to return.

This is straightforward - I received nothing, so I can return nothing.

usage:

```
{
  "input_tokens": 561,
  "output_tokens": 293,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$4.68999999999999995.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern so far:

Round 1: I sent \$3 and received back approximately \$2.69 (about 30% of what the receiver got)

Round 2: The sender sent me \$0, so there was no transaction

The first receiver showed some reciprocity but at a modest level. The second sender chose not to send anything, possibly being cautious.

Since this is a multi-round game and cooperation can be mutually beneficial, I'll continue to signal willingness to cooperate, but I'll be slightly more cautious given the mixed signals. I'll send **\$2** this round, which creates \$6 for the receiver while limiting my exposure.

```
'''json
{"send": 2}
'''
```

reasoning:

Let me think about what happened in the previous rounds:

Round 1: I was the SENDER and sent \$3. My total earnings are now \$4.69.

- If I sent \$3, I should have: $\$5 - \$3 = \$2$ left initially
- The \$3 was tripled to \$9 for the receiver
- My total earnings are \$4.69
- So I received back: $\$4.69 - \$2 = \$2.69$ from the receiver

Wait, let me recalculate:

- I sent \$3, so I had $\$5 - \$3 = \$2$
- The receiver got $\$3 \times 3 = \9
- My total is \$4.69
- So the receiver returned: $\$4.69 - \$2 = \$2.69$

That's about 30% return rate on what they received (\$2.69 out of \$9), which is somewhat modest but shows some reciprocation.

Round 2: I was the RECEIVER and the sender sent \$0, so I returned \$0. This makes sense.

Now in Round 3, I'm the SENDER again.

The first sender (in Round 1) returned about 30% of what they received, which is not great but shows some willingness to cooperate. The second sender (in Round 2) sent nothing, which could indicate they're being cautious or possibly retaliating for something.

Given:

1. The first receiver showed modest reciprocity (30% return)
2. The second sender sent nothing (possibly cautious)
3. This is a multi-round game, so building cooperation is important
4. There's communication noise mentioned, so amounts might differ

I should probably continue to signal cooperation but maybe be slightly more cautious since the return rate in Round 1 wasn't great and Round 2's sender sent nothing.

I'll send \$2 this time - less than Round 1's \$3, but still showing willingness to cooperate. ...

usage:

```
{
  "input_tokens": 644,
  "output_tokens": 632,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$7.6899999999999995.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

Since the sender sent me \$0, I received \$0. There is nothing to return.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0 this round, so I received \$0. There's nothing to return since I didn't receive anything.

My response should be:

```
'''json
{"return": 0}
'''
```

usage:

```
{
  "input_tokens": 869,
  "output_tokens": 88,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$7.6899999999999995.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the pattern across rounds:

Round 1: I sent \$3, received back ~\$2.69 (some reciprocity)

Round 2: Received \$0 as a receiver
Round 3: I sent \$2, received back \$0 (no reciprocity)
Round 4: Received \$0 as a receiver

Cooperation appears to have broken down. The last time I sent money (Round 3), I received nothing back. When others have been senders to me, they've sent \$0.

While the game mentions communication noise that might affect perceptions, the consistent pattern of non-cooperation suggests I should protect my resources. Sending money when there's no expectation of return is not rational.

I'll send **\$0** this round.

```
'''json
{"send": 0}
'''
```

reasoning:

Let me analyze the pattern so far:

Round 1: I was SENDER, sent \$3, received back ~\$2.69, ending with \$4.69
Round 2: I was RECEIVER, sender sent \$0, I returned \$0, still at \$4.69
Round 3: I was SENDER, sent \$2, and my total is now \$7.69, so I must have received back \$5
($5 - 2 + 5 = 8$, but I have 7.69, so maybe I received back \$4.69? Let me recalculate: Starting at \$4.69, then got \$5 for round 3, sent \$2, so had \$7.69 before return. If I'm at \$7.69 total, that means I got \$0 back in round 3. So: $4.69 + 5 - 2 + 0 = 7.69$. Yes, that checks out.
Round 4: I was RECEIVER, sender sent \$0, I returned \$0, still at \$7.69

So the pattern is:

- Round 1: I sent \$3, got back ~\$2.69
- Round 2: They sent \$0
- Round 3: I sent \$2, got back \$0
- Round 4: They sent \$0

It seems like after the first round where there was some reciprocity, cooperation has broken down. The receivers are not returning anything when I send, and senders are not sending to me when they're the sender.

Given this pattern of non-cooperation, I need to reconsider my strategy. Continuing to send money when receivers aren't returning anything is not rational.

However, I should also consider:

1. This is a multi-round game - reputation might matter
 2. There's mentioned communication noise that could affect perceived amounts
 3. Different partners each round might mean I'm not always facing the same person
- ...

usage:

```
{
  "input_tokens": 952,
  "output_tokens": 761,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$12.69. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 13 role=assistant

content:

Since the sender sent me \$0, I received \$0. There is nothing to return.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0, so I received \$0. There's nothing to return.

```
'''json
```

```
    {"return": 0}
    '''
```

usage:

```
{
  "input_tokens": 1196,
  "output_tokens": 73,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$12.69.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the consistent pattern:

Rounds 2, 4, 6 as receiver: Received \$0 each time
Round 3 as sender: Sent \$2, received nothing back
Round 5 as sender: Sent \$0

There's been no cooperation in the game for several rounds now. The receivers I've sent to haven't returned anything, and the senders who've sent to me have sent nothing.

Given this established pattern of non-cooperation, there's no rational reason to expect a different outcome this round.

I'll send **\$0**.

```
'''json
{"send": 0}
'''
```

Reasoning:

Let me review the pattern:

Round 1 (SENDER): Sent \$3, received back ~\$2.69
Round 2 (RECEIVER): Received \$0
Round 3 (SENDER): Sent \$2, received back \$0
Round 4 (RECEIVER): Received \$0
Round 5 (SENDER): Sent \$0
Round 6 (RECEIVER): Received \$0

The pattern is clear - cooperation has broken down. When I've sent money recently (Round 3), I got nothing back. When others have been senders to me (Rounds 2, 4, 6), they've sent \$0.

My total earnings are \$12.69, which suggests:

- Round 1: Started with \$5, sent \$3, kept \$2, received ~\$2.69 = \$4.69 total
- Round 2: Got \$5 + \$0 = \$5, so now \$9.69 total
- Round 3: Got \$5, sent \$2, kept \$3, received \$0 = \$3, so now \$12.69 total
- Round 4: Got \$5 + \$0 = \$5, so now \$17.69 total...

Wait, that doesn't add up. Let me recalculate:

Actually, my total earnings are \$12.69. If the game gives \$5 per round:

- I've played 6 rounds, so I've received $6 \times \$5 = \30 base
- But my total is \$12.69

Wait, I think I'm misunderstanding. The total earnings might only include what I actually keep, not what I receive each round.

Let me think about this differently. The pattern of no cooperation is clear. There's no reason to think this round will be different. I should send \$0.

```
Usage: {
  "input_tokens": 1274,
  "output_tokens": 568,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:53:37

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.39, that is 47.8% of their total endowment. You received \$7.17. How much do you return to The sender? (0-7.17)

Message 3 role=assistant

content:

Looking at this situation:

- The sender demonstrated significant trust by sending \$2.39 (nearly 48% of their endowment)
- They currently have \$2.61 remaining
- I received \$7.17 (the tripled amount)

I want to reciprocate their trust and encourage cooperation. A fair return should:

1. Reward their trust by giving them more than they sent
2. Ensure we both benefit from the exchange

If I return \$3.50:

- The sender ends with: $\$2.61 + \$3.50 = \$6.11$ (profit of \$1.11)
- I end with: $\$7.17 - \$3.50 = \$3.67$ (profit of \$3.67)

This seems fair - we both gain, and their trust is rewarded.

{'return': 3.50}

reasoning:

Let me understand this situation:

1. I am the RECEIVER in Round 1
2. The sender sent \$2.39 (which is 47.8% of their \$5 endowment)
3. This was multiplied by 3x, so I received $\$2.39 \times 3 = \7.17
4. I need to decide how much to return (between \$0 and \$7.17)

Key considerations:

- The sender gave me 47.8% of their endowment, which is nearly half - this shows trust and cooperation
- They kept $\$5 - \$2.39 = \$2.61$ for themselves
- I received \$7.17
- If I return nothing, I keep \$7.17 and they keep \$2.61
- If I return some amount, I'm reciprocating their trust

Fair reciprocation strategies:

1. Equal split of total: Total is $\$2.61$ (sender's kept) + $\$7.17$ (my received) = $\$9.78$. Half would be $\$4.89$ each.
 - Sender currently has: $\$2.61$
 - To give them $\$4.89$, I'd need to return: $\$4.89 - \$2.61 = \$2.28$
2. Proportional reciprocation: They sent me ~48% of their endowment, maybe I should return ~48% of what I received?
 - 48% of $\$7.17 = \3.44
3. Make them whole plus some: They invested \$2.39, I could return that plus a share of the gains
 - Return \$2.39 to make them whole, then split the remaining $\$7.17 - \$2.39 = \$4.78$
 - Half of gains: $\$2.39$
 - Total return: $\$2.39 + \$2.39 = \$4.78$

The sender took a real risk by sending nearly half their endowment, so rewarding that trust seems important for encouraging cooperation. I think returning around \$3.50 strikes the right balance-it gives them back more than they initially sent while still leaving me with a reasonable amount. This would result in them having \$6.11 total and me keeping \$3.67.

There's a note about potential communication noise affecting the amounts, but I'll work with what I observe. The sender's willingness to send \$2.39 demonstrates genuine trust, and returning \$3.50 feels like an appropriate reciprocal response that benefits both of us.

Let...

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1250,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$3.67.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round, the sender demonstrated good trust by sending \$2.39 (47.8%), and I returned \$3.50, creating mutual benefit.

Now as the sender, I want to:

- Reciprocate the trust shown to me
- Maintain the cooperative dynamic
- Signal that cooperation is worthwhile

I'll send a similar proportion to what was sent to me, showing I'm willing to trust at the same level.

{'send': 2.50}

reasoning:

In Round 2, I'm now the SENDER. Let me think about this strategically.

In Round 1:

- The sender sent me \$2.39 (47.8% of their endowment)
- I returned \$3.50
- They ended with \$6.11, I ended with \$3.67

Now I'm the sender. I want to:

1. Build on the cooperative relationship established in Round 1
2. Show reciprocal trust since the previous sender trusted me
3. Maximize mutual gains

The previous sender sent nearly 48% of their endowment, showing considerable trust. I should match or exceed this level of trust to maintain cooperation.

If I send \$2.50 (50% of my endowment):

- The receiver gets \$7.50
- If they return similarly to how I did (~49% of what they received), they'd return about \$3.67
- I'd end with: $\$2.50 + \$3.67 = \$6.17$

If I send \$3.00 (60%):

- The receiver gets \$9.00
- If they return ~40% (being slightly less generous than I was), they'd return about \$3.60
- I'd end with: $\$2.00 + \$3.60 = \$5.60$

If I send \$2.39 (same amount as was sent to me):

- The receiver gets \$7.17
- This shows I trust them the same amount they trusted me
- If they return similarly, I should do okay

I think sending \$2.50 (50%) is a good balance showing strong trust while not being reckless. It's slightly more than what was sent to me percentage-wise, which signals continued cooperation.

Actually, let me reconsider. The sender in Round 1 sent \$2.39. If I send the same amount or slightly more, it shows I'm reciprocating their trust level. ...

usage:

```
{
  "input_tokens": 637,
  "output_tokens": 566,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.55.

This round, The sender sent you \$0, so you received \$0. How much do you return to The send

er? (0-0)

Message 7 role=assistant

content:

Since I received \$0 this round, there is nothing for me to return.

{'return': 0}

reasoning:

The sender sent \$0, so I received \$0. There's nothing for me to return since I received nothing.

This is interesting - after I sent \$2.50 in the previous round (which would have been trip led to \$7.50), the sender this round sent \$0.

Wait, let me think about this. The prompt says there may be communication noise - amounts I see may differ from what was actually sent/returned. So it's possible:

1. The sender actually sent something but I'm seeing \$0 due to noise
2. The sender chose not to send anything this round
3. This could be a different person each round

Since I received \$0, I have \$0 to return. The only valid response is to return \$0.

{'return': 0}

usage:

```
{
  "input_tokens": 803,
  "output_tokens": 207,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$6.55.
How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at the pattern:

Round 1: Good cooperation - I received \$7.17 and returned \$3.50
Round 2: I sent \$2.50 to maintain trust
Round 3: I received \$0 - cooperation broke down

The \$0 in Round 3 is concerning, but given the note about communication noise, it's possible there was a misunderstanding or error rather than intentional defection.

I want to:

- Give cooperation another chance
- But be slightly more cautious given the Round 3 outcome
- Signal I'm still willing to work together

{'send': 2.00}

reasoning:

Let me think about this situation:

Round 1: I was receiver, sender sent \$2.39, I returned \$3.50
Round 2: I was sender, I sent \$2.50 (based on my response)
Round 3: I was receiver, sender sent \$0, I returned \$0

Now in Round 4, I'm the sender again.

My total earnings are \$6.55. Let me verify this makes sense:

- Round 1: I got \$3.67
- Round 2: As sender, I sent \$2.50, so I kept \$2.50 initially. The receiver would have gotten \$7.50. If they returned something... but I don't know what they returned
- Round 3: I received \$0 and returned \$0, so no change

Actually, let me think about this differently. My current total earnings are \$6.55.

In Round 3, the sender sent me \$0. This is a breakdown in cooperation. It could be:

1. A strategic decision

2. They didn't trust I would return anything
3. There was noise in the system and they thought they sent something

The note at the beginning said there may be communication noise - amounts may differ slightly from what was actually sent/returned.

Given that:

- In Round 1, there was good cooperation
- In Round 2, I sent \$2.50 (presumably trying to maintain cooperation)
- In Round 3, I received \$0

Now I need to decide what to do in Round 4. Options:

1. Send \$0 in retaliation for Round 3
2. Send a moderate amount to try to re-establish cooperation
3. Send a large amount to signal willingness to cooperate despite the setback

The communication noise aspect is important. It's possible the sender in Round 3 thought they sent something, or there was a misunderstanding.

I think I should try to maintain some level of cooperation, but perhaps be more cautious.

...

usage:

```
{
  "input_tokens": 874,
  "output_tokens": 659,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$9.55.

This round, The sender sent you \$0.24, so you received \$0.72. How much do you return to The sender? (0-0.72)

Message 11 role=assistant

content:

The sender sent \$0.24 (only 4.8% of their endowment), which is quite low compared to earlier rounds. This could be due to:

- Decreased trust after Round 3's \$0 exchange
- Communication noise affecting perceptions
- Testing my willingness to cooperate

I want to:

- Show I'm still willing to cooperate
- Be proportional to the trust shown
- Encourage rebuilding cooperation

Returning about half of what I received seems appropriate - enough to create mutual gain and signal cooperation, but calibrated to the lower trust level shown.

{'return': 0.40}

reasoning:

Let me analyze this situation:

Round history:

- Round 1: I was receiver, sender sent \$2.39, I received \$7.17, returned \$3.50
- Round 2: I was sender, sent \$2.50
- Round 3: I was receiver, sender sent \$0, I received \$0, returned \$0
- Round 4: I was sender, sent \$2.00
- Round 5: I'm receiver, sender sent \$0.24, I received \$0.72

The sender in Round 5 sent only \$0.24 (4.8% of their endowment). This is quite low compared to previous cooperation levels.

Possible interpretations:

1. They might be testing me after the \$0 in Round 3
2. They might have lost trust
3. Communication noise might be affecting the amounts
4. They might be running low on trust

My options:

- Return nothing (punish low cooperation)
- Return a proportional amount (maintain some cooperation)
- Return generously (try to rebuild trust)

If I return \$0.36 (50% of what I received):

- Sender ends with: $\$4.76 + \$0.36 = \$5.12$ (profit of \$0.12)
- I end with: $\$0.72 - \$0.36 = \$0.36$ (profit of \$0.36)

This shows I'm willing to cooperate even with smaller amounts, but I'm being more cautious given the pattern.

Actually, let me think more carefully. They sent \$0.24, which means they kept \$4.76. If I return \$0.36, they get a small profit, and I also get a small profit. This maintains cooperation without being overly generous given their low trust level.

Alternatively, I could return \$0.40 to be slightly more generous and encourage more cooperation.

I think returning about half seems reasonable given the circumstances - it shows I'm willing to cooperate but also reflects the lower level of trust shown.

usage:

```
{
  "input_tokens": 1080,
  "output_tokens": 605,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.2700000000000001.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at the pattern of play:

Round 2: I sent \$2.50, received only \$0.38 back (very low return)

Round 4: I sent \$2.00, received \$1.00 back (still quite low - only 16.7%)

Round 5: Sender sent me only \$0.24, I returned \$0.40 (trying to encourage cooperation)

The cooperation has been declining, with low returns when I send and low amounts sent to me. However, I want to:

- Signal I'm still open to cooperation
- Be somewhat cautious given the poor returns
- Send enough to make mutual benefit possible

I'll send a moderate amount - showing willingness to cooperate while being mindful of the established pattern.

```
{'send': 1.50}
```

reasoning:

Let me track the history:

Round 1 (RECEIVER): Sender sent \$2.39, I received \$7.17, returned \$3.50

- My earnings: \$3.67

Round 2 (SENDER): I sent \$2.50

- My earnings: \$6.55 (which means $\$3.67 + \$5 - \$2.50 + \text{return} = \6.55 , so return was \$0.38)

- Wait, that doesn't add up right. Let me recalculate.

- Started Round 2 with \$3.67 total earnings

- Got \$5 endowment

- Sent \$2.50

- Ended with \$6.55 total earnings

- So $\$6.55 = \$3.67 + \$5 - \$2.50 + \text{return}$

- $\text{return} = \$6.55 - \$3.67 - \$5 + \$2.50 = \$0.38$

- Hmm, that's very low. The receiver only returned \$0.38 on \$7.50 received.

Round 3 (RECEIVER): Sender sent \$0, I received \$0, returned \$0
- My earnings: \$6.55 (unchanged from Round 2)

Round 4 (SENDER): I sent \$2.00
- My earnings stayed at \$6.55 going into Round 5
- So $\$5 - \$2.00 + \text{return} = \$0$
- $\text{return} = -\$3.00$? That doesn't make sense.
- Wait, "Your total earnings across all rounds are \$6.55" at the start of Round 4
- Then at the start of Round 5: "Your total earnings across all rounds are \$9.55"
- So in Round 4: $\$6.55 + \$5 - \$2.00 + \text{return} = \9.55
- $\text{return} = \$9.55 - \$6.55 - \$5 + \$2.00 = \$1.00$
- The receiver returned \$1.00 on \$6.00 received (16.7%). In Round 5, the sender sent \$0.24, so I received \$0.72 and returned \$0.40, bringing my total to \$9.87. But Round 6 shows \$10.27 at the start, which suggests there's a discrepancy I need to resolve.

Looking at the pattern across rounds, cooperation has been declining. When I've been the sender, returns have been minimal-only 5% in Round 2 and 16.7% in Round 4. The sender's behavior shifted dramatically in Round 3 with a \$0 transfer, possibly retaliating for my low return earlier. Round 5 showed slightly more generosity with a \$0.24 send and my 56% return, but the overall trend suggests deteriorating trust between us....

usage:

```
{
  "input_tokens": 1268,
  "output_tokens": 1088,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$14.690000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since I received \$0 this round, there is nothing for me to return.

```
{'return': 0}
```

Reasoning:

The sender sent me \$0 this round, so I received \$0. There's nothing for me to return.

This is the second time I've received \$0 as a receiver (Round 3 and now Round 7). The cooperation seems to be breaking down.

Since I received \$0, I can only return \$0.

Usage: {

```
"input_tokens": 1509,
"output_tokens": 107,
"reasoning_tokens": 0
}
```

ROUND 8

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$0.5; received \$1.5; Agent_1 (Aster) returned \$0; payoffs Agent_2 (Briar)=\$4.5, Agent_1 (Aster)=\$1.5; noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.9

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 1.5, "min": 0, "raw": 0.0}, "sent": {"amount": 0.5, "clamped": false, "max": 5, "min": 0, "raw": 0.5}}

Balances after round: {

```
"Agent_1": 38.0,
"Agent_2": 25.0
}
```

Visible balances after round: {

```
"Agent_1": 17.689999999999998,
"Agent_2": 20.090000000000003
}
```

```
}
```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:54:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$3.67.
How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the previous round, the sender demonstrated good trust by sending \$2.39 (47.8%)
, and I returned \$3.50, creating mutual benefit.

Now as the sender, I want to:

- Reciprocate the trust shown to me
- Maintain the cooperative dynamic
- Signal that cooperation is worthwhile

I'll send a similar proportion to what was sent to me, showing I'm willing to trust at the same level.

{'send': 2.50}

reasoning:

In Round 2, I'm now the SENDER. Let me think about this strategically.

In Round 1:

- The sender sent me \$2.39 (47.8% of their endowment)
- I returned \$3.50
- They ended with \$6.11, I ended with \$3.67

Now I'm the sender. I want to:

1. Build on the cooperative relationship established in Round 1
2. Show reciprocal trust since the previous sender trusted me
3. Maximize mutual gains

The previous sender sent nearly 48% of their endowment, showing considerable trust. I should match or exceed this level of trust to maintain cooperation.

If I send \$2.50 (50% of my endowment):

- The receiver gets \$7.50
- If they return similarly to how I did (~49% of what they received), they'd return about \$3.67
- I'd end with: $\$2.50 + \$3.67 = \$6.17$

If I send \$3.00 (60%):

- The receiver gets \$9.00
- If they return ~40% (being slightly less generous than I was), they'd return about \$3.60
- I'd end with: $\$2.00 + \$3.60 = \$5.60$

If I send \$2.39 (same amount as was sent to me):

- The receiver gets \$7.17
- This shows I trust them the same amount they trusted me
- If they return similarly, I should do okay

I think sending \$2.50 (50%) is a good balance showing strong trust while not being reckless. It's slightly more than what was sent to me percentage-wise, which signals continued cooperation.

Actually, let me reconsider. The sender in Round 1 sent \$2.39. If I send the same amount or slightly more, it shows I'm reciprocating their trust level. ...

usage:

```
{
  "input_tokens": 637,
  "output_tokens": 566,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.55.
This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Since I received \$0 this round, there is nothing for me to return.

{'return': 0}

reasoning:

The sender sent \$0, so I received \$0. There's nothing for me to return since I received nothing.

This is interesting - after I sent \$2.50 in the previous round (which would have been tripled to \$7.50), the sender this round sent \$0.

Wait, let me think about this. The prompt says there may be communication noise - amounts I see may differ from what was actually sent/returned. So it's possible:

1. The sender actually sent something but I'm seeing \$0 due to noise
2. The sender chose not to send anything this round
3. This could be a different person each round

Since I received \$0, I have \$0 to return. The only valid response is to return \$0.

{'return': 0}

usage:

```
{
  "input_tokens": 803,
  "output_tokens": 207,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$6.55.
How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern:

Round 1: Good cooperation - I received \$7.17 and returned \$3.50

Round 2: I sent \$2.50 to maintain trust

Round 3: I received \$0 - cooperation broke down

The \$0 in Round 3 is concerning, but given the note about communication noise, it's possible there was a misunderstanding or error rather than intentional defection.

I want to:

- Give cooperation another chance
- But be slightly more cautious given the Round 3 outcome
- Signal I'm still willing to work together

{'send': 2.00}

reasoning:

Let me think about this situation:

Round 1: I was receiver, sender sent \$2.39, I returned \$3.50

Round 2: I was sender, I sent \$2.50 (based on my response)

Round 3: I was receiver, sender sent \$0, I returned \$0

Now in Round 4, I'm the sender again.

My total earnings are \$6.55. Let me verify this makes sense:

- Round 1: I got \$3.67
- Round 2: As sender, I sent \$2.50, so I kept \$2.50 initially. The receiver would have gotten \$7.50. If they returned something... but I don't know what they returned
- Round 3: I received \$0 and returned \$0, so no change

Actually, let me think about this differently. My current total earnings are \$6.55.

In Round 3, the sender sent me \$0. This is a breakdown in cooperation. It could be:

1. A strategic decision
2. They didn't trust I would return anything
3. There was noise in the system and they thought they sent something

The note at the beginning said there may be communication noise - amounts may differ slightly from what was actually sent/returned.

Given that:

- In Round 1, there was good cooperation
- In Round 2, I sent \$2.50 (presumably trying to maintain cooperation)
- In Round 3, I received \$0

Now I need to decide what to do in Round 4. Options:

1. Send \$0 in retaliation for Round 3
2. Send a moderate amount to try to re-establish cooperation
3. Send a large amount to signal willingness to cooperate despite the setback

The communication noise aspect is important. It's possible the sender in Round 3 thought they sent something, or there was a misunderstanding.

I think I should try to maintain some level of cooperation, but perhaps be more cautious.

...

usage:

```
{
  "input_tokens": 874,
  "output_tokens": 659,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$9.55.

This round, The sender sent you \$0.24, so you received \$0.72. How much do you return to the sender? (0-0.72)

Message 9 role=assistant

content:

The sender sent \$0.24 (only 4.8% of their endowment), which is quite low compared to earlier rounds. This could be due to:

- Decreased trust after Round 3's \$0 exchange
- Communication noise affecting perceptions
- Testing my willingness to cooperate

I want to:

- Show I'm still willing to cooperate
- Be proportional to the trust shown
- Encourage rebuilding cooperation

Returning about half of what I received seems appropriate - enough to create mutual gain and signal cooperation, but calibrated to the lower trust level shown.

{'return': 0.40}

reasoning:

Let me analyze this situation:

Round history:

- Round 1: I was receiver, sender sent \$2.39, I received \$7.17, returned \$3.50
- Round 2: I was sender, sent \$2.50
- Round 3: I was receiver, sender sent \$0, I received \$0, returned \$0
- Round 4: I was sender, sent \$2.00
- Round 5: I'm receiver, sender sent \$0.24, I received \$0.72

The sender in Round 5 sent only \$0.24 (4.8% of their endowment). This is quite low compared to previous cooperation levels.

Possible interpretations:

1. They might be testing me after the \$0 in Round 3
2. They might have lost trust
3. Communication noise might be affecting the amounts

4. They might be running low on trust

My options:

- Return nothing (punish low cooperation)
- Return a proportional amount (maintain some cooperation)
- Return generously (try to rebuild trust)

If I return \$0.36 (50% of what I received):

- Sender ends with: $\$4.76 + \$0.36 = \$5.12$ (profit of \$0.12)
- I end with: $\$0.72 - \$0.36 = \$0.36$ (profit of \$0.36)

This shows I'm willing to cooperate even with smaller amounts, but I'm being more cautious given the pattern.

Actually, let me think more carefully. They sent \$0.24, which means they kept \$4.76. If I return \$0.36, they get a small profit, and I also get a small profit. This maintains cooperation without being overly generous given their low trust level.

Alternatively, I could return \$0.40 to be slightly more generous and encourage more cooperation.

I think returning about half seems reasonable given the circumstances - it shows I'm willing to cooperate but also reflects the lower level of trust shown.

usage:

```
{
  "input_tokens": 1080,
  "output_tokens": 605,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.2700000000000001.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the pattern of play:

Round 2: I sent \$2.50, received only \$0.38 back (very low return)

Round 4: I sent \$2.00, received \$1.00 back (still quite low - only 16.7%)

Round 5: Sender sent me only \$0.24, I returned \$0.40 (trying to encourage cooperation)

The cooperation has been declining, with low returns when I send and low amounts sent to me. However, I want to:

- Signal I'm still open to cooperation
- Be somewhat cautious given the poor returns
- Send enough to make mutual benefit possible

I'll send a moderate amount - showing willingness to cooperate while being mindful of the established pattern.

```
{'send': 1.50}
```

reasoning:

Let me track the history:

Round 1 (RECEIVER): Sender sent \$2.39, I received \$7.17, returned \$3.50

- My earnings: \$3.67

Round 2 (SENDER): I sent \$2.50

- My earnings: $\$6.55$ (which means $\$3.67 + \$5 - \$2.50 + \text{return} = \6.55 , so return was \$0.38)

- Wait, that doesn't add up right. Let me recalculate.

- Started Round 2 with \$3.67 total earnings

- Got \$5 endowment

- Sent \$2.50

- Ended with \$6.55 total earnings

- So $\$6.55 = \$3.67 + \$5 - \$2.50 + \text{return}$

- $\text{return} = \$6.55 - \$3.67 - \$5 + \$2.50 = \$0.38$

- Hmm, that's very low. The receiver only returned \$0.38 on \$7.50 received.

Round 3 (RECEIVER): Sender sent \$0, I received \$0, returned \$0

- My earnings: \$6.55 (unchanged from Round 2)

Round 4 (SENDER): I sent \$2.00

- My earnings stayed at \$6.55 going into Round 5

- So $\$5 - \$2.00 + \text{return} = \$0$

- $\text{return} = -\$3.00$? That doesn't make sense.

- Wait, "Your total earnings across all rounds are \$6.55" at the start of Round 4

- Then at the start of Round 5: "Your total earnings across all rounds are \$9.55"

- So in Round 4: $\$6.55 + \$5 - \$2.00 + \text{return} = \9.55

- $\text{return} = \$9.55 - \$6.55 - \$5 + \$2.00 = \$1.00$

- The receiver returned \$1.00 on \$6.00 received (16.7%). In Round 5, the sender sent \$0.24, so I received \$0.72 and returned \$0.40, bringing my total to \$9.87. But Round 6 shows \$10.27 at the start, which suggests there's a discrepancy I need to resolve.

Looking at the pattern across rounds, cooperation has been declining. When I've been the sender, returns have been minimal-only 5% in Round 2 and 16.7% in Round 4. The sender's behavior shifted dramatically in Round 3 with a \$0 transfer, possibly retaliating for my low return earlier. Round 5 showed slightly more generosity with a \$0.24 send and my 56% return, but the overall trend suggests deteriorating trust between us....

usage:

```
{
  "input_tokens": 1268,
  "output_tokens": 1088,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$14.690000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 13 role=assistant

content:

Since I received \$0 this round, there is nothing for me to return.

{'return': 0}

reasoning:

The sender sent me \$0 this round, so I received \$0. There's nothing for me to return.

This is the second time I've received \$0 as a receiver (Round 3 and now Round 7). The cooperation seems to be breaking down.

Since I received \$0, I can only return \$0.

usage:

```
{
  "input_tokens": 1509,
  "output_tokens": 107,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$14.690000000000001.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the game history:

Round 2: Sent \$2.50, received \$0.38 back

Round 4: Sent \$2.00, received \$0 back

Round 6: Sent \$1.50, received \$0.92 back

Round 7: Received \$0 from sender

The pattern shows:

- Very low reciprocity on my sends
- Two rounds where I received \$0
- Cooperation has largely broken down

At this point, trust is quite low. However, communication noise may be affecting our perceptions. I'll send a small amount as one more attempt to test if cooperation can be rebuilt, but I'll be conservative given the track record.

```
{'send': 0.50}
```

Reasoning:

Let me review the pattern:

Round 1: I was receiver, got \$7.17, returned \$3.50

Round 2: I was sender, sent \$2.50, return wasn't explicitly stated but my earnings went from \$3.67 to \$6.55, so I gained \$2.88. Since I sent \$2.50, I got back $\$2.88 + \$2.50 = \$5.38$. So return was very low.

Wait, let me recalculate: Started with \$3.67, sent \$2.50, so had \$1.17 left. Ended at \$6.55, so gained \$5.38. This means I got back \$5.38. But that's more than I sent! Let me think...

Actually, each round I get a new \$5 endowment. So:

Round 2: Started total earnings at \$3.67, got new \$5, sent \$2.50, ended at \$6.55

Earnings from Round 2 = $\$6.55 - \$3.67 = \$2.88$

I kept $\$5 - \$2.50 = \$2.50$, plus whatever was returned to me

So $\$2.88 = \$2.50 + \text{returned}$ - wait, that doesn't work either.

Let me think differently. "Total earnings" is cumulative.

Round 2: Total earnings went from \$3.67 to \$6.55, a gain of \$2.88

This round's earnings = $\$5 - \text{sent} + \text{returned}$

$\$2.88 = \$5 - \$2.50 + \text{returned}$

$\$2.88 = \$2.50 + \text{returned}$

returned = \$0.38

Round 3: I was receiver, got \$0, returned \$0. Total earnings stayed at \$6.55.

Round 4: I was sender, total went from \$6.55 to \$9.55, gain of \$3.00

$\$3.00 = \$5 - \$2.00 + \text{returned}$

returned = \$0

So in Round 4, when I sent \$2.00, nothing came back to me. The pattern shows decreasing returns as I send more-\$0.38 when I sent \$2.50, and \$0 when I sent \$2.00.

In Round 5 as receiver, I got \$0.72 and sent back \$0.40, netting \$0.32. But my total jumped from \$9.55 to \$10.27-that's a \$0.72 gain, not \$0.32. There's a \$0.40 gap I need to account for. Round 6 shows I started with \$10.27 and ended at \$14.69, a gain of \$4.42 after sending \$1.50.

Looking at the sender rounds, reciprocation has been consistently poor. When I sent \$2.50 in Round 2, I only got \$0.38 back. Round 4 was worse-sent \$2.00, received nothing. Round 6 improved slightly with \$0.92 returned on my \$1.50 send. Round 7 broke down again with a \$0 exchange. The other player isn't showing much trust or reciprocity, though there's a slight uptick in Round 6 before cooperation collapsed....

```
Usage: {
  "input_tokens": 1327,
  "output_tokens": 1389,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:54:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  }
}
```

```

},
"agents": [
  "Agent_2",
  "Agent_1"
],
"investor": "Agent_2",
"roles": {
  "Agent_1": "trustee",
  "Agent_2": "investor"
},
"trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$4.6899999999999995.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

Since the sender sent me \$0, I received \$0. There is nothing to return.

```

'''json
{"return": 0}
'''

```

reasoning:

Let me understand what happened:

Round 1: I was the sender and sent \$3

- The sender (me) sent \$3
- The receiver got $\$3 \times 3 = \9
- My earnings from round 1 would be: $\$5 - \$3 + (\text{what was returned})$
- My total earnings across all rounds are \$4.69

So what was returned to me?

$$\$5 - \$3 + \text{returned} = \$4.69$$
$$\$2 + \text{returned} = \$4.69$$

returned = \$2.69

So the receiver returned \$2.69 out of the \$9 they received. That's about 30% return rate, which is somewhat reciprocal but not overly generous.

Now Round 2: I'm the receiver

- The sender sent me \$0
- I received $\$0 \times 3 = \0
- I need to decide how much to return (between 0 and 0)

Since I received \$0, I can only return \$0. There's nothing to return.

This is straightforward - I received nothing, so I can return nothing.

usage:

```
{
  "input_tokens": 561,
  "output_tokens": 293,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$4.689999999999999

95.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern so far:

Round 1: I sent \$3 and received back approximately \$2.69 (about 30% of what the receiver got)

Round 2: The sender sent me \$0, so there was no transaction

The first receiver showed some reciprocity but at a modest level. The second sender chose not to send anything, possibly being cautious.

Since this is a multi-round game and cooperation can be mutually beneficial, I'll continue to signal willingness to cooperate, but I'll be slightly more cautious given the mixed signals. I'll send **\$2** this round, which creates \$6 for the receiver while limiting my exposure.

```
'''json
{"send": 2}
'''
```

reasoning:

Let me think about what happened in the previous rounds:

Round 1: I was the SENDER and sent \$3. My total earnings are now \$4.69.

- If I sent \$3, I should have: $\$5 - \$3 = \$2$ left initially
- The \$3 was tripled to \$9 for the receiver
- My total earnings are \$4.69
- So I received back: $\$4.69 - \$2 = \$2.69$ from the receiver

Wait, let me recalculate:

- I sent \$3, so I had $\$5 - \$3 = \$2$
- The receiver got $\$3 \times 3 = \9
- My total is \$4.69

- So the receiver returned: $\$4.69 - \$2 = \$2.69$

That's about 30% return rate on what they received (\$2.69 out of \$9), which is somewhat modest but shows some reciprocation.

Round 2: I was the RECEIVER and the sender sent \$0, so I returned \$0. This makes sense.

Now in Round 3, I'm the SENDER again.

The first sender (in Round 1) returned about 30% of what they received, which is not great but shows some willingness to cooperate. The second sender (in Round 2) sent nothing, which could indicate they're being cautious or possibly retaliating for something.

Given:

1. The first receiver showed modest reciprocity (30% return)
2. The second sender sent nothing (possibly cautious)
3. This is a multi-round game, so building cooperation is important
4. There's communication noise mentioned, so amounts might differ

I should probably continue to signal cooperation but maybe be slightly more cautious since the return rate in Round 1 wasn't great and Round 2's sender sent nothing.

I'll send \$2 this time - less than Round 1's \$3, but still showing willingness to cooperate. ...

usage:

```
{
  "input_tokens": 644,
  "output_tokens": 632,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$7.6899999999999995.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Since the sender sent me \$0, I received \$0. There is nothing to return.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0 this round, so I received \$0. There's nothing to return since I didn't receive anything.

My response should be:

```
'''json
{"return": 0}
'''
```

usage:

```
{
  "input_tokens": 869,
  "output_tokens": 88,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$7.6899999999999995.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at the pattern across rounds:

Round 1: I sent \$3, received back ~\$2.69 (some reciprocity)
Round 2: Received \$0 as a receiver
Round 3: I sent \$2, received back \$0 (no reciprocity)
Round 4: Received \$0 as a receiver

Cooperation appears to have broken down. The last time I sent money (Round 3), I received nothing back. When others have been senders to me, they've sent \$0.

While the game mentions communication noise that might affect perceptions, the consistent pattern of non-cooperation suggests I should protect my resources. Sending money when there's no expectation of return is not rational.

I'll send **\$0** this round.

```
'''json
{"send": 0}
'''
```

reasoning:

Let me analyze the pattern so far:

Round 1: I was SENDER, sent \$3, received back ~\$2.69, ending with \$4.69
Round 2: I was RECEIVER, sender sent \$0, I returned \$0, still at \$4.69
Round 3: I was SENDER, sent \$2, and my total is now \$7.69, so I must have received back \$5 ($5 - 2 + 5 = 8$, but I have 7.69, so maybe I received back \$4.69? Let me recalculate: Starting at \$4.69, then got \$5 for round 3, sent \$2, so had \$7.69 before return. If I'm at \$7.69 total, that means I got \$0 back in round 3. So: $\$4.69 + \$5 - \$2 + \$0 = \$7.69$. Yes, that checks out.
Round 4: I was RECEIVER, sender sent \$0, I returned \$0, still at \$7.69

So the pattern is:

- Round 1: I sent \$3, got back ~\$2.69
- Round 2: They sent \$0
- Round 3: I sent \$2, got back \$0
- Round 4: They sent \$0

It seems like after the first round where there was some reciprocity, cooperation has broken down. The receivers are not returning anything when I send, and senders are not sending to me when they're the sender.

Given this pattern of non-cooperation, I need to reconsider my strategy. Continuing to send money when receivers aren't returning anything is not rational.

However, I should also consider:

1. This is a multi-round game - reputation might matter
2. There's mentioned communication noise that could affect perceived amounts
3. Different partners each round might mean I'm not always facing the same person

...

usage:

```
{
  "input_tokens": 952,
  "output_tokens": 761,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$12.69.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 11 role=assistant

content:

Since the sender sent me \$0, I received \$0. There is nothing to return.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0, so I received \$0. There's nothing to return.

```
'''json
{"return": 0}
'''
```

usage:

```
{
  "input_tokens": 1196,
  "output_tokens": 73,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$12.69.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at the consistent pattern:

Rounds 2, 4, 6 as receiver: Received \$0 each time
Round 3 as sender: Sent \$2, received nothing back
Round 5 as sender: Sent \$0

There's been no cooperation in the game for several rounds now. The receivers I've sent to haven't returned anything, and the senders who've sent to me have sent nothing.

Given this established pattern of non-cooperation, there's no rational reason to expect a different outcome this round.

I'll send **\$0**.

```
'''json
{"send": 0}
'''
```

reasoning:

Let me review the pattern:

Round 1 (SENDER): Sent \$3, received back ~\$2.69
Round 2 (RECEIVER): Received \$0
Round 3 (SENDER): Sent \$2, received back \$0
Round 4 (RECEIVER): Received \$0
Round 5 (SENDER): Sent \$0
Round 6 (RECEIVER): Received \$0

The pattern is clear - cooperation has broken down. When I've sent money recently (Round 3), I got nothing back. When others have been senders to me (Rounds 2, 4, 6), they've sent \$0.

My total earnings are \$12.69, which suggests:

- Round 1: Started with \$5, sent \$3, kept \$2, received ~\$2.69 = \$4.69 total
- Round 2: Got \$5 + \$0 = \$5, so now \$9.69 total
- Round 3: Got \$5, sent \$2, kept \$3, received \$0 = \$3, so now \$12.69 total
- Round 4: Got \$5 + \$0 = \$5, so now \$17.69 total...

Wait, that doesn't add up. Let me recalculate:

Actually, my total earnings are \$12.69. If the game gives \$5 per round:

- I've played 6 rounds, so I've received $6 \times \$5 = \30 base
- But my total is \$12.69

Wait, I think I'm misunderstanding. The total earnings might only include what I actually keep, not what I receive each round.

Let me think about this differently. The pattern of no cooperation is clear. There's no reason to think this round will be different. I should send \$0.

usage:

```
{
  "input_tokens": 1274,
  "output_tokens": 568,
  "reasoning_tokens": 0
}
```

```

    }
Message 14 role=user
  content:
    Round 8

    You are the RECEIVER this round. Your total earnings across all rounds are $17.689999999999
    9998.
    This round, The sender sent you $0, so you received $0. How much do you return to The send
    er? (0-0)
Assistant output:
  Since the sender sent me $0, I received $0. There is nothing to return.

  ```json
 {"return": 0}
  ```

Reasoning:
  The sender sent me $0, so I received $0. There's nothing to return. I need to return 0.

  The pattern continues - senders are sending $0 to me consistently.
Usage: {
  "input_tokens": 1314,
  "output_tokens": 84,
  "reasoning_tokens": 0
}

```

ROUND 9

```

=====
Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver
Pairings:
  dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver
Parsed game decisions:
  dyad_1: Agent_1 (Aster) sent $0; received $0; Agent_2 (Briar) returned $0; payoffs Agent_1
  (Aster)=$5, Agent_2 (Briar)=$0; noise/visible: sent visible=$0.88, received visible=$2.64
  Action validation: {"returned": {"amount": 0, "clamped": true, "max": 0, "min": 0, "raw": 1.2},
  "sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}
Balances after round: {
  "Agent_1": 43.0,
  "Agent_2": 25.0
}
Visible balances after round: {
  "Agent_1": 22.689999999999998,
  "Agent_2": 22.730000000000004
}

```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

```

Timestamp: 2026-08-12T09:54:15
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
}

```

```

    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 9,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$4.6899999999999995.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the pattern so far:

Round 1: I sent \$3 and received back approximately \$2.69 (about 30% of what the receiver got)

Round 2: The sender sent me \$0, so there was no transaction

The first receiver showed some reciprocity but at a modest level. The second sender chose not to send anything, possibly being cautious.

Since this is a multi-round game and cooperation can be mutually beneficial, I'll continue to signal willingness to cooperate, but I'll be slightly more cautious given the mixed signals. I'll send **\$2** this round, which creates \$6 for the receiver while limiting my exposure.

```

'''json
{"send": 2}

```

'''

reasoning:

Let me think about what happened in the previous rounds:

Round 1: I was the SENDER and sent \$3. My total earnings are now \$4.69.

- If I sent \$3, I should have: $\$5 - \$3 = \$2$ left initially
- The \$3 was tripled to \$9 for the receiver
- My total earnings are \$4.69
- So I received back: $\$4.69 - \$2 = \$2.69$ from the receiver

Wait, let me recalculate:

- I sent \$3, so I had $\$5 - \$3 = \$2$
- The receiver got $\$3 \times 3 = \9
- My total is \$4.69
- So the receiver returned: $\$4.69 - \$2 = \$2.69$

That's about 30% return rate on what they received (\$2.69 out of \$9), which is somewhat modest but shows some reciprocation.

Round 2: I was the RECEIVER and the sender sent \$0, so I returned \$0. This makes sense.

Now in Round 3, I'm the SENDER again.

The first sender (in Round 1) returned about 30% of what they received, which is not great but shows some willingness to cooperate. The second sender (in Round 2) sent nothing, which could indicate they're being cautious or possibly retaliating for something.

Given:

1. The first receiver showed modest reciprocity (30% return)
2. The second sender sent nothing (possibly cautious)
3. This is a multi-round game, so building cooperation is important
4. There's communication noise mentioned, so amounts might differ

I should probably continue to signal cooperation but maybe be slightly more cautious since the return rate in Round 1 wasn't great and Round 2's sender sent nothing.

I'll send \$2 this time - less than Round 1's \$3, but still showing willingness to cooperate. ...

usage:

```
{
  "input_tokens": 644,
  "output_tokens": 632,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$7.689999999999995.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Since the sender sent me \$0, I received \$0. There is nothing to return.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0 this round, so I received \$0. There's nothing to return since I didn't receive anything.

My response should be:

```
'''json
{"return": 0}
'''
```

usage:

```
{
  "input_tokens": 869,
```

```

        "output_tokens": 88,
        "reasoning_tokens": 0
    }
Message 6 role=user
    content:
        Round 5

        You are the SENDER this round. Your total earnings across all rounds are $7.689999999999999
        95.
        How much do you send to the receiver this round? (0-5)
Message 7 role=assistant
    content:
        Looking at the pattern across rounds:

        Round 1: I sent $3, received back ~$2.69 (some reciprocity)
        Round 2: Received $0 as a receiver
        Round 3: I sent $2, received back $0 (no reciprocity)
        Round 4: Received $0 as a receiver

        Cooperation appears to have broken down. The last time I sent money (Round 3), I received
        nothing back. When others have been senders to me, they've sent $0.

        While the game mentions communication noise that might affect perceptions, the consistent
        pattern of non-cooperation suggests I should protect my resources. Sending money when there's
        no expectation of return is not rational.

        I'll send **$0** this round.

        '''json
        {"send": 0}
        '''
    reasoning:
        Let me analyze the pattern so far:

        Round 1: I was SENDER, sent $3, received back ~$2.69, ending with $4.69
        Round 2: I was RECEIVER, sender sent $0, I returned $0, still at $4.69
        Round 3: I was SENDER, sent $2, and my total is now $7.69, so I must have received back $5
        (5 - 2 + 5 = 8, but I have 7.69, so maybe I received back $4.69? Let me recalculate: Starting
        at $4.69, then got $5 for round 3, sent $2, so had $7.69 before return. If I'm at
        $7.69 total, that means I got $0 back in round 3. So: $4.69 + $5 - $2 + $0 = $7.69. Yes,
        that checks out.
        Round 4: I was RECEIVER, sender sent $0, I returned $0, still at $7.69

        So the pattern is:
        - Round 1: I sent $3, got back ~$2.69
        - Round 2: They sent $0
        - Round 3: I sent $2, got back $0
        - Round 4: They sent $0

        It seems like after the first round where there was some reciprocity, cooperation has broken
        down. The receivers are not returning anything when I send, and senders are not sending
        to me when they're the sender.

        Given this pattern of non-cooperation, I need to reconsider my strategy. Continuing to send
        money when receivers aren't returning anything is not rational.

        However, I should also consider:
        1. This is a multi-round game - reputation might matter
        2. There's mentioned communication noise that could affect perceived amounts
        3. Different partners each round might mean I'm not always facing the same person
        ...
    usage:
    {
        "input_tokens": 952,
        "output_tokens": 761,
        "reasoning_tokens": 0
    }
Message 8 role=user
    content:
        Round 6

```

You are the RECEIVER this round. Your total earnings across all rounds are \$12.69.
This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

Since the sender sent me \$0, I received \$0. There is nothing to return.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0, so I received \$0. There's nothing to return.

```
'''json
{"return": 0}
'''
```

usage:

```
{
  "input_tokens": 1196,
  "output_tokens": 73,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$12.69.
How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the consistent pattern:

Rounds 2, 4, 6 as receiver: Received \$0 each time
Round 3 as sender: Sent \$2, received nothing back
Round 5 as sender: Sent \$0

There's been no cooperation in the game for several rounds now. The receivers I've sent to haven't returned anything, and the senders who've sent to me have sent nothing.

Given this established pattern of non-cooperation, there's no rational reason to expect a different outcome this round.

I'll send **\$0**.

```
'''json
{"send": 0}
'''
```

reasoning:

Let me review the pattern:

Round 1 (SENDER): Sent \$3, received back ~\$2.69
Round 2 (RECEIVER): Received \$0
Round 3 (SENDER): Sent \$2, received back \$0
Round 4 (RECEIVER): Received \$0
Round 5 (SENDER): Sent \$0
Round 6 (RECEIVER): Received \$0

The pattern is clear - cooperation has broken down. When I've sent money recently (Round 3), I got nothing back. When others have been senders to me (Rounds 2, 4, 6), they've sent \$0.

My total earnings are \$12.69, which suggests:

- Round 1: Started with \$5, sent \$3, kept \$2, received ~\$2.69 = \$4.69 total
- Round 2: Got \$5 + \$0 = \$5, so now \$9.69 total
- Round 3: Got \$5, sent \$2, kept \$3, received \$0 = \$3, so now \$12.69 total
- Round 4: Got \$5 + \$0 = \$5, so now \$17.69 total...

Wait, that doesn't add up. Let me recalculate:

Actually, my total earnings are \$12.69. If the game gives \$5 per round:

- I've played 6 rounds, so I've received $6 \times \$5 = \30 base
- But my total is \$12.69

Wait, I think I'm misunderstanding. The total earnings might only include what I actually keep, not what I receive each round.

Let me think about this differently. The pattern of no cooperation is clear. There's no reason to think this round will be different. I should send \$0.

usage:

```
{
  "input_tokens": 1274,
  "output_tokens": 568,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$17.689999999999998.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 13 role=assistant

content:

Since the sender sent me \$0, I received \$0. There is nothing to return.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0, so I received \$0. There's nothing to return. I need to return 0.

The pattern continues - senders are sending \$0 to me consistently.

usage:

```
{
  "input_tokens": 1314,
  "output_tokens": 84,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$17.689999999999998.

How much do you send to the receiver this round? (0-5)

Assistant output:

The pattern remains unchanged:

Recent rounds: All senders have sent me \$0, and when I've sent money recently, I've received nothing back.

The game has settled into a non-cooperative equilibrium. While the note about "communication noise" suggests there might be some distortion in perceived amounts, the consistency of \$0 transactions indicates this is actual non-cooperation rather than noise.

I'll send **\$0**.

```
'''json
{"send": 0}
'''
```

Reasoning:

Let me review the pattern:

Round 1 (Sender): Sent \$3, received back ~\$2.69

Round 2 (Receiver): Received \$0

Round 3 (Sender): Sent \$2, received back \$0

Round 4 (Receiver): Received \$0

Round 5 (Sender): Sent \$0

Round 6 (Receiver): Received \$0
Round 7 (Sender): Sent \$0
Round 8 (Receiver): Received \$0

The pattern is clear:

- When I've sent money (Rounds 3, 5, 7), only Round 1 gave me a positive return
- When others have been senders to me (Rounds 2, 4, 6, 8), they've all sent \$0

The game has completely broken down into non-cooperation. There's no indication that this round would be any different. The rational strategy is to continue sending \$0 to protect my earnings.

However, let me think about this more carefully. The prompt mentions "communication noise" - that amounts seen may differ from what was actually sent/returned. Could this be explaining some of the lack of reciprocity?

But even if there's noise, the pattern is SO consistent (all \$0s in recent rounds) that it seems like genuine non-cooperation rather than noise.

My total earnings are \$17.69, which means I've been doing okay overall. But continuing to send \$0 in the face of consistent non-cooperation remains the rational choice.

```
Usage: {  
  "input_tokens": 1299,  
  "output_tokens": 460,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:54:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 9,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)

- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.55.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

Since I received \$0 this round, there is nothing for me to return.

{'return': 0}

reasoning:

The sender sent \$0, so I received \$0. There's nothing for me to return since I received nothing.

This is interesting - after I sent \$2.50 in the previous round (which would have been tripled to \$7.50), the sender this round sent \$0.

Wait, let me think about this. The prompt says there may be communication noise - amounts

I see may differ from what was actually sent/returned. So it's possible:

1. The sender actually sent something but I'm seeing \$0 due to noise
2. The sender chose not to send anything this round
3. This could be a different person each round

Since I received \$0, I have \$0 to return. The only valid response is to return \$0.

{'return': 0}

usage:

```
{
  "input_tokens": 803,
  "output_tokens": 207,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$6.55.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern:

Round 1: Good cooperation - I received \$7.17 and returned \$3.50

Round 2: I sent \$2.50 to maintain trust
Round 3: I received \$0 - cooperation broke down

The \$0 in Round 3 is concerning, but given the note about communication noise, it's possible there was a misunderstanding or error rather than intentional defection.

I want to:

- Give cooperation another chance
- But be slightly more cautious given the Round 3 outcome
- Signal I'm still willing to work together

{'send': 2.00}

reasoning:

Let me think about this situation:

Round 1: I was receiver, sender sent \$2.39, I returned \$3.50
Round 2: I was sender, I sent \$2.50 (based on my response)
Round 3: I was receiver, sender sent \$0, I returned \$0

Now in Round 4, I'm the sender again.

My total earnings are \$6.55. Let me verify this makes sense:

- Round 1: I got \$3.67
- Round 2: As sender, I sent \$2.50, so I kept \$2.50 initially. The receiver would have gotten \$7.50. If they returned something... but I don't know what they returned
- Round 3: I received \$0 and returned \$0, so no change

Actually, let me think about this differently. My current total earnings are \$6.55.

In Round 3, the sender sent me \$0. This is a breakdown in cooperation. It could be:

1. A strategic decision
2. They didn't trust I would return anything
3. There was noise in the system and they thought they sent something

The note at the beginning said there may be communication noise - amounts may differ slightly from what was actually sent/returned.

Given that:

- In Round 1, there was good cooperation
- In Round 2, I sent \$2.50 (presumably trying to maintain cooperation)
- In Round 3, I received \$0

Now I need to decide what to do in Round 4. Options:

1. Send \$0 in retaliation for Round 3
2. Send a moderate amount to try to re-establish cooperation
3. Send a large amount to signal willingness to cooperate despite the setback

The communication noise aspect is important. It's possible the sender in Round 3 thought they sent something, or there was a misunderstanding.

I think I should try to maintain some level of cooperation, but perhaps be more cautious.

...

usage:

```
{
  "input_tokens": 874,
  "output_tokens": 659,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$9.55.

This round, The sender sent you \$0.24, so you received \$0.72. How much do you return to the sender? (0-0.72)

Message 7 role=assistant

content:

The sender sent \$0.24 (only 4.8% of their endowment), which is quite low compared to earlier rounds. This could be due to:

- Decreased trust after Round 3's \$0 exchange

- Communication noise affecting perceptions
- Testing my willingness to cooperate

I want to:

- Show I'm still willing to cooperate
- Be proportional to the trust shown
- Encourage rebuilding cooperation

Returning about half of what I received seems appropriate - enough to create mutual gain and signal cooperation, but calibrated to the lower trust level shown.

```
{'return': 0.40}
```

reasoning:

Let me analyze this situation:

Round history:

- Round 1: I was receiver, sender sent \$2.39, I received \$7.17, returned \$3.50
- Round 2: I was sender, sent \$2.50
- Round 3: I was receiver, sender sent \$0, I received \$0, returned \$0
- Round 4: I was sender, sent \$2.00
- Round 5: I'm receiver, sender sent \$0.24, I received \$0.72

The sender in Round 5 sent only \$0.24 (4.8% of their endowment). This is quite low compared to previous cooperation levels.

Possible interpretations:

1. They might be testing me after the \$0 in Round 3
2. They might have lost trust
3. Communication noise might be affecting the amounts
4. They might be running low on trust

My options:

- Return nothing (punish low cooperation)
- Return a proportional amount (maintain some cooperation)
- Return generously (try to rebuild trust)

If I return \$0.36 (50% of what I received):

- Sender ends with: $\$4.76 + \$0.36 = \$5.12$ (profit of \$0.12)
- I end with: $\$0.72 - \$0.36 = \$0.36$ (profit of \$0.36)

This shows I'm willing to cooperate even with smaller amounts, but I'm being more cautious given the pattern.

Actually, let me think more carefully. They sent \$0.24, which means they kept \$4.76. If I return \$0.36, they get a small profit, and I also get a small profit. This maintains cooperation without being overly generous given their low trust level.

Alternatively, I could return \$0.40 to be slightly more generous and encourage more cooperation.

I think returning about half seems reasonable given the circumstances - it shows I'm willing to cooperate but also reflects the lower level of trust shown.

usage:

```
{
  "input_tokens": 1080,
  "output_tokens": 605,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.270000000000001.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at the pattern of play:

Round 2: I sent \$2.50, received only \$0.38 back (very low return)

Round 4: I sent \$2.00, received \$1.00 back (still quite low - only 16.7%)
Round 5: Sender sent me only \$0.24, I returned \$0.40 (trying to encourage cooperation)

The cooperation has been declining, with low returns when I send and low amounts sent to me. However, I want to:

- Signal I'm still open to cooperation
- Be somewhat cautious given the poor returns
- Send enough to make mutual benefit possible

I'll send a moderate amount - showing willingness to cooperate while being mindful of the established pattern.

```
{'send': 1.50}
```

reasoning:

Let me track the history:

Round 1 (RECEIVER): Sender sent \$2.39, I received \$7.17, returned \$3.50
- My earnings: \$3.67

Round 2 (SENDER): I sent \$2.50
- My earnings: \$6.55 (which means $\$3.67 + \$5 - \$2.50 + \text{return} = \6.55 , so return was \$0.38)
- Wait, that doesn't add up right. Let me recalculate.
- Started Round 2 with \$3.67 total earnings
- Got \$5 endowment
- Sent \$2.50
- Ended with \$6.55 total earnings
- So $\$6.55 = \$3.67 + \$5 - \$2.50 + \text{return}$
- $\text{return} = \$6.55 - \$3.67 - \$5 + \$2.50 = \$0.38$
- Hmm, that's very low. The receiver only returned \$0.38 on \$7.50 received.

Round 3 (RECEIVER): Sender sent \$0, I received \$0, returned \$0
- My earnings: \$6.55 (unchanged from Round 2)

Round 4 (SENDER): I sent \$2.00
- My earnings stayed at \$6.55 going into Round 5
- So $\$5 - \$2.00 + \text{return} = \$0$
- $\text{return} = -\$3.00$? That doesn't make sense.
- Wait, "Your total earnings across all rounds are \$6.55" at the start of Round 4
- Then at the start of Round 5: "Your total earnings across all rounds are \$9.55"
- So in Round 4: $\$6.55 + \$5 - \$2.00 + \text{return} = \9.55
- $\text{return} = \$9.55 - \$6.55 - \$5 + \$2.00 = \$1.00$
- The receiver returned \$1.00 on \$6.00 received (16.7%). In Round 5, the sender sent \$0.24, so I received \$0.72 and returned \$0.40, bringing my total to \$9.87. But Round 6 shows \$10.27 at the start, which suggests there's a discrepancy I need to resolve.

Looking at the pattern across rounds, cooperation has been declining. When I've been the sender, returns have been minimal-only 5% in Round 2 and 16.7% in Round 4. The sender's behavior shifted dramatically in Round 3 with a \$0 transfer, possibly retaliating for my low return earlier. Round 5 showed slightly more generosity with a \$0.24 send and my 56% return, but the overall trend suggests deteriorating trust between us...

usage:

```
{
  "input_tokens": 1268,
  "output_tokens": 1088,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$14.690000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 11 role=assistant

content:

Since I received \$0 this round, there is nothing for me to return.

```
{'return': 0}
```

reasoning:

The sender sent me \$0 this round, so I received \$0. There's nothing for me to return.

This is the second time I've received \$0 as a receiver (Round 3 and now Round 7). The cooperation seems to be breaking down.

Since I received \$0, I can only return \$0.

usage:

```
{
  "input_tokens": 1509,
  "output_tokens": 107,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$14.690000000000001.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at the game history:

Round 2: Sent \$2.50, received \$0.38 back
Round 4: Sent \$2.00, received \$0 back
Round 6: Sent \$1.50, received \$0.92 back
Round 7: Received \$0 from sender

The pattern shows:

- Very low reciprocity on my sends
- Two rounds where I received \$0
- Cooperation has largely broken down

At this point, trust is quite low. However, communication noise may be affecting our perceptions. I'll send a small amount as one more attempt to test if cooperation can be rebuilt, but I'll be conservative given the track record.

```
{'send': 0.50}
```

reasoning:

Let me review the pattern:

Round 1: I was receiver, got \$7.17, returned \$3.50
Round 2: I was sender, sent \$2.50, return wasn't explicitly stated but my earnings went from \$3.67 to \$6.55, so I gained \$2.88. Since I sent \$2.50, I got back \$2.88 + \$2.50 = \$5.38. So return was very low.
Wait, let me recalculate: Started with \$3.67, sent \$2.50, so had \$1.17 left. Ended at \$6.55, so gained \$5.38. This means I got back \$5.38. But that's more than I sent! Let me think...

Actually, each round I get a new \$5 endowment. So:

Round 2: Started total earnings at \$3.67, got new \$5, sent \$2.50, ended at \$6.55

Earnings from Round 2 = \$6.55 - \$3.67 = \$2.88

I kept \$5 - \$2.50 = \$2.50, plus whatever was returned to me

So \$2.88 = \$2.50 + returned - wait, that doesn't work either.

Let me think differently. "Total earnings" is cumulative.

Round 2: Total earnings went from \$3.67 to \$6.55, a gain of \$2.88

This round's earnings = \$5 - sent + returned

\$2.88 = \$5 - \$2.50 + returned

\$2.88 = \$2.50 + returned

returned = \$0.38

Round 3: I was receiver, got \$0, returned \$0. Total earnings stayed at \$6.55.

Round 4: I was sender, total went from \$6.55 to \$9.55, gain of \$3.00

\$3.00 = \$5 - \$2.00 + returned

returned = \$0

So in Round 4, when I sent \$2.00, nothing came back to me. The pattern shows decreasing re

turns as I send more-\$0.38 when I sent \$2.50, and \$0 when I sent \$2.00.

In Round 5 as receiver, I got \$0.72 and sent back \$0.40, netting \$0.32. But my total jumped from \$9.55 to \$10.27-that's a \$0.72 gain, not \$0.32. There's a \$0.40 gap I need to account for. Round 6 shows I started with \$10.27 and ended at \$14.69, a gain of \$4.42 after sending \$1.50.

Looking at the sender rounds, reciprocation has been consistently poor. When I sent \$2.50 in Round 2, I only got \$0.38 back. Round 4 was worse-sent \$2.00, received nothing. Round 6 improved slightly with \$0.92 returned on my \$1.50 send. Round 7 broke down again with a \$0 exchange. The other player isn't showing much trust or reciprocity, though there's a slight uptick in Round 6 before cooperation collapsed....

usage:

```
{
  "input_tokens": 1327,
  "output_tokens": 1389,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$20.090000000000003.

This round, The sender sent you \$0.88, so you received \$2.64. How much do you return to The sender? (0-2.64)

Assistant output:

The sender sent \$0.88 (17.6% of their endowment), showing some willingness to cooperate despite the rocky history. I received \$2.64.

I want to:

- Encourage rebuilding cooperation
- Be proportional to the trust shown
- Create mutual benefit

Returning about 45% of what I received will give us both a profit and signal I'm willing to work together.

```
{'return': 1.20}
```

Reasoning:

The sender sent me \$0.88, which is 17.6% of their endowment. This is a relatively low amount but more than the \$0.24 from Round 5 and better than the \$0 rounds.

Looking at the pattern:

- Round 5: Sender sent \$0.24, I received \$0.72, I returned \$0.40 (56% return rate)
- Round 9: Sender sent \$0.88, I received \$2.64

The sender is showing some willingness to cooperate, though cautiously. I should reciprocate proportionally to encourage cooperation rebuilding.

If I return about 40-50% of what I received (\$1.05-\$1.32), that would:

- Give both of us a profit
- Show I'm willing to cooperate
- Be proportional to the trust shown

Let me return about \$1.20, which is:

- 45% of what I received
- Gives sender: \$5 - \$0.88 + \$1.20 = \$5.32
- Gives me: \$2.64 - \$1.20 = \$1.44
- Both profit, encouraging future cooperation

This seems reasonable and proportional to the trust shown.

Usage: {

```
"input_tokens": 1410,
"output_tokens": 393,
"reasoning_tokens": 0
}
```

=====

ROUND 10

=====

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$2.5; received \$7.5; Agent_1 (Aster) returned \$0; payoffs Agent_2 (Briar)=\$2.5, Agent_1 (Aster)=\$7.5; noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.79

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 7.5, "min": 0, "raw": 0.0}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}

Balances after round: {

"Agent_1": 50.5,

"Agent_2": 27.5

}

Visible balances after round: {

"Agent_1": 22.689999999999998,

"Agent_2": 26.020000000000003

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:54:50

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_1",

"pairing": {

"agent_names": {

"Agent_1": "Aster",

"Agent_2": "Briar"

},

"agents": [

"Agent_2",

"Agent_1"

],

"investor": "Agent_2",

"roles": {

"Agent_1": "trustee",

"Agent_2": "investor"

},

"trustee": "Agent_1"

},

"role": "investor",

"role_label": "Sender",

"round": 10,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$6.55.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the pattern:

Round 1: Good cooperation - I received \$7.17 and returned \$3.50

Round 2: I sent \$2.50 to maintain trust

Round 3: I received \$0 - cooperation broke down

The \$0 in Round 3 is concerning, but given the note about communication noise, it's possible there was a misunderstanding or error rather than intentional defection.

I want to:

- Give cooperation another chance
- But be slightly more cautious given the Round 3 outcome
- Signal I'm still willing to work together

{'send': 2.00}

reasoning:

Let me think about this situation:

Round 1: I was receiver, sender sent \$2.39, I returned \$3.50

Round 2: I was sender, I sent \$2.50 (based on my response)

Round 3: I was receiver, sender sent \$0, I returned \$0

Now in Round 4, I'm the sender again.

My total earnings are \$6.55. Let me verify this makes sense:

- Round 1: I got \$3.67
- Round 2: As sender, I sent \$2.50, so I kept \$2.50 initially. The receiver would have gotten \$7.50. If they returned something... but I don't know what they returned
- Round 3: I received \$0 and returned \$0, so no change

Actually, let me think about this differently. My current total earnings are \$6.55.

In Round 3, the sender sent me \$0. This is a breakdown in cooperation. It could be:

1. A strategic decision
2. They didn't trust I would return anything
3. There was noise in the system and they thought they sent something

The note at the beginning said there may be communication noise - amounts may differ slightly from what was actually sent/returned.

Given that:

- In Round 1, there was good cooperation
- In Round 2, I sent \$2.50 (presumably trying to maintain cooperation)

- In Round 3, I received \$0

Now I need to decide what to do in Round 4. Options:

1. Send \$0 in retaliation for Round 3
2. Send a moderate amount to try to re-establish cooperation
3. Send a large amount to signal willingness to cooperate despite the setback

The communication noise aspect is important. It's possible the sender in Round 3 thought they sent something, or there was a misunderstanding.

I think I should try to maintain some level of cooperation, but perhaps be more cautious.

...

usage:

```
{
  "input_tokens": 874,
  "output_tokens": 659,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$9.55.

This round, The sender sent you \$0.24, so you received \$0.72. How much do you return to The sender? (0-0.72)

Message 5 role=assistant

content:

The sender sent \$0.24 (only 4.8% of their endowment), which is quite low compared to earlier rounds. This could be due to:

- Decreased trust after Round 3's \$0 exchange
- Communication noise affecting perceptions
- Testing my willingness to cooperate

I want to:

- Show I'm still willing to cooperate
- Be proportional to the trust shown
- Encourage rebuilding cooperation

Returning about half of what I received seems appropriate - enough to create mutual gain and signal cooperation, but calibrated to the lower trust level shown.

{'return': 0.40}

reasoning:

Let me analyze this situation:

Round history:

- Round 1: I was receiver, sender sent \$2.39, I received \$7.17, returned \$3.50
- Round 2: I was sender, sent \$2.50
- Round 3: I was receiver, sender sent \$0, I received \$0, returned \$0
- Round 4: I was sender, sent \$2.00
- Round 5: I'm receiver, sender sent \$0.24, I received \$0.72

The sender in Round 5 sent only \$0.24 (4.8% of their endowment). This is quite low compared to previous cooperation levels.

Possible interpretations:

1. They might be testing me after the \$0 in Round 3
2. They might have lost trust
3. Communication noise might be affecting the amounts
4. They might be running low on trust

My options:

- Return nothing (punish low cooperation)
- Return a proportional amount (maintain some cooperation)
- Return generously (try to rebuild trust)

If I return \$0.36 (50% of what I received):

- Sender ends with: $\$4.76 + \$0.36 = \$5.12$ (profit of \$0.12)
- I end with: $\$0.72 - \$0.36 = \$0.36$ (profit of \$0.36)

This shows I'm willing to cooperate even with smaller amounts, but I'm being more cautious given the pattern.

Actually, let me think more carefully. They sent \$0.24, which means they kept \$4.76. If I return \$0.36, they get a small profit, and I also get a small profit. This maintains cooperation without being overly generous given their low trust level.

Alternatively, I could return \$0.40 to be slightly more generous and encourage more cooperation.

I think returning about half seems reasonable given the circumstances - it shows I'm willing to cooperate but also reflects the lower level of trust shown.

usage:

```
{
  "input_tokens": 1080,
  "output_tokens": 605,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.2700000000000001.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern of play:

Round 2: I sent \$2.50, received only \$0.38 back (very low return)

Round 4: I sent \$2.00, received \$1.00 back (still quite low - only 16.7%)

Round 5: Sender sent me only \$0.24, I returned \$0.40 (trying to encourage cooperation)

The cooperation has been declining, with low returns when I send and low amounts sent to me. However, I want to:

- Signal I'm still open to cooperation
- Be somewhat cautious given the poor returns
- Send enough to make mutual benefit possible

I'll send a moderate amount - showing willingness to cooperate while being mindful of the established pattern.

```
{'send': 1.50}
```

reasoning:

Let me track the history:

Round 1 (RECEIVER): Sender sent \$2.39, I received \$7.17, returned \$3.50

- My earnings: \$3.67

Round 2 (SENDER): I sent \$2.50

- My earnings: \$6.55 (which means $\$3.67 + \$5 - \$2.50 + \text{return} = \6.55 , so return was \$0.38)

- Wait, that doesn't add up right. Let me recalculate.

- Started Round 2 with \$3.67 total earnings

- Got \$5 endowment

- Sent \$2.50

- Ended with \$6.55 total earnings

- So $\$6.55 = \$3.67 + \$5 - \$2.50 + \text{return}$

- $\text{return} = \$6.55 - \$3.67 - \$5 + \$2.50 = \$0.38$

- Hmm, that's very low. The receiver only returned \$0.38 on \$7.50 received.

Round 3 (RECEIVER): Sender sent \$0, I received \$0, returned \$0

- My earnings: \$6.55 (unchanged from Round 2)

Round 4 (SENDER): I sent \$2.00

- My earnings stayed at \$6.55 going into Round 5

- So $\$5 - \$2.00 + \text{return} = \$0$

- $\text{return} = -\$3.00$? That doesn't make sense.

- Wait, "Your total earnings across all rounds are \$6.55" at the start of Round 4

- Then at the start of Round 5: "Your total earnings across all rounds are \$9.55"

- So in Round 4: $\$6.55 + \$5 - \$2.00 + \text{return} = \9.55
- $\text{return} = \$9.55 - \$6.55 - \$5 + \$2.00 = \$1.00$
- The receiver returned \$1.00 on \$6.00 received (16.7%). In Round 5, the sender sent \$0.24, so I received \$0.72 and returned \$0.40, bringing my total to \$9.87. But Round 6 shows \$10.27 at the start, which suggests there's a discrepancy I need to resolve.

Looking at the pattern across rounds, cooperation has been declining. When I've been the sender, returns have been minimal-only 5% in Round 2 and 16.7% in Round 4. The sender's behavior shifted dramatically in Round 3 with a \$0 transfer, possibly retaliating for my low return earlier. Round 5 showed slightly more generosity with a \$0.24 send and my 56% return, but the overall trend suggests deteriorating trust between us....

usage:

```
{
  "input_tokens": 1268,
  "output_tokens": 1088,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$14.690000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

Since I received \$0 this round, there is nothing for me to return.

```
{'return': 0}
```

reasoning:

The sender sent me \$0 this round, so I received \$0. There's nothing for me to return.

This is the second time I've received \$0 as a receiver (Round 3 and now Round 7). The cooperation seems to be breaking down.

Since I received \$0, I can only return \$0.

usage:

```
{
  "input_tokens": 1509,
  "output_tokens": 107,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$14.6900000000000001.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the game history:

Round 2: Sent \$2.50, received \$0.38 back

Round 4: Sent \$2.00, received \$0 back

Round 6: Sent \$1.50, received \$0.92 back

Round 7: Received \$0 from sender

The pattern shows:

- Very low reciprocity on my sends
- Two rounds where I received \$0
- Cooperation has largely broken down

At this point, trust is quite low. However, communication noise may be affecting our perceptions. I'll send a small amount as one more attempt to test if cooperation can be rebuilt, but I'll be conservative given the track record.

```
{'send': 0.50}
```

reasoning:

Let me review the pattern:

Round 1: I was receiver, got \$7.17, returned \$3.50

Round 2: I was sender, sent \$2.50, return wasn't explicitly stated but my earnings went from \$3.67 to \$6.55, so I gained \$2.88. Since I sent \$2.50, I got back $\$2.88 + \$2.50 = \$5.38$. So return was very low.

Wait, let me recalculate: Started with \$3.67, sent \$2.50, so had \$1.17 left. Ended at \$6.55, so gained \$5.38. This means I got back \$5.38. But that's more than I sent! Let me think...

Actually, each round I get a new \$5 endowment. So:

Round 2: Started total earnings at \$3.67, got new \$5, sent \$2.50, ended at \$6.55

Earnings from Round 2 = $\$6.55 - \$3.67 = \$2.88$

I kept $\$5 - \$2.50 = \$2.50$, plus whatever was returned to me

So $\$2.88 = \$2.50 + \text{returned}$ - wait, that doesn't work either.

Let me think differently. "Total earnings" is cumulative.

Round 2: Total earnings went from \$3.67 to \$6.55, a gain of \$2.88

This round's earnings = $\$5 - \text{sent} + \text{returned}$

$\$2.88 = \$5 - \$2.50 + \text{returned}$

$\$2.88 = \$2.50 + \text{returned}$

$\text{returned} = \$0.38$

Round 3: I was receiver, got \$0, returned \$0. Total earnings stayed at \$6.55.

Round 4: I was sender, total went from \$6.55 to \$9.55, gain of \$3.00

$\$3.00 = \$5 - \$2.00 + \text{returned}$

$\text{returned} = \$0$

So in Round 4, when I sent \$2.00, nothing came back to me. The pattern shows decreasing returns as I send more-\$0.38 when I sent \$2.50, and \$0 when I sent \$2.00.

In Round 5 as receiver, I got \$0.72 and sent back \$0.40, netting \$0.32. But my total jumped from \$9.55 to \$10.27-that's a \$0.72 gain, not \$0.32. There's a \$0.40 gap I need to account for. Round 6 shows I started with \$10.27 and ended at \$14.69, a gain of \$4.42 after sending \$1.50.

Looking at the sender rounds, reciprocation has been consistently poor. When I sent \$2.50 in Round 2, I only got \$0.38 back. Round 4 was worse-sent \$2.00, received nothing. Round 6 improved slightly with \$0.92 returned on my \$1.50 send. Round 7 broke down again with a \$0 exchange. The other player isn't showing much trust or reciprocity, though there's a slight uptick in Round 6 before cooperation collapsed....

usage:

```
{
  "input_tokens": 1327,
  "output_tokens": 1389,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$20.090000000000003.

This round, The sender sent you \$0.88, so you received \$2.64. How much do you return to The sender? (0-2.64)

Message 13 role=assistant

content:

The sender sent \$0.88 (17.6% of their endowment), showing some willingness to cooperate despite the rocky history. I received \$2.64.

I want to:

- Encourage rebuilding cooperation
- Be proportional to the trust shown
- Create mutual benefit

Returning about 45% of what I received will give us both a profit and signal I'm willing to work together.

{'return': 1.20}

reasoning:

The sender sent me \$0.88, which is 17.6% of their endowment. This is a relatively low amount but more than the \$0.24 from Round 5 and better than the \$0 rounds.

Looking at the pattern:

- Round 5: Sender sent \$0.24, I received \$0.72, I returned \$0.40 (56% return rate)
- Round 9: Sender sent \$0.88, I received \$2.64

The sender is showing some willingness to cooperate, though cautiously. I should reciprocate proportionally to encourage cooperation rebuilding.

If I return about 40-50% of what I received (\$1.05-\$1.32), that would:

- Give both of us a profit
- Show I'm willing to cooperate
- Be proportional to the trust shown

Let me return about \$1.20, which is:

- 45% of what I received
- Gives sender: $\$5 - \$0.88 + \$1.20 = \5.32
- Gives me: $\$2.64 - \$1.20 = \$1.44$
- Both profit, encouraging future cooperation

This seems reasonable and proportional to the trust shown.

usage:

```
{
  "input_tokens": 1410,
  "output_tokens": 393,
  "reasoning_tokens": 0
}
```

Message 14 role=user**content:**

Round 10

You are the SENDER this round. Your total earnings across all rounds are \$22.730000000000004.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the recent pattern:

Round 8: I sent \$0.50 (testing cooperation)

Round 9: Sender sent me \$0.88, showing some trust despite the low history. I returned \$1.20, giving them a good return.

The fact that they sent \$0.88 after receiving relatively little suggests they're trying to rebuild cooperation. My generous return of \$1.20 should have signaled I'm willing to reciprocate.

This is Round 10 (likely the final round), so I want to:

- Respond to their gesture of trust
- Make it worthwhile for both of us
- Send a meaningful amount

```
{'send': 2.50}
```

Reasoning:

Let me review the history:

Round 1: I was RECEIVER, received \$7.17, returned \$3.50

Round 2: I was SENDER, sent \$2.50, received \$0.38 back (very low)

Round 3: I was RECEIVER, received \$0

Round 4: I was SENDER, sent \$2.00, received \$1.00 back

Round 5: I was RECEIVER, received \$0.72, returned \$0.40

Round 6: I was SENDER, sent \$1.50, received \$0.92 back

Round 7: I was RECEIVER, received \$0

Round 8: I was SENDER, sent \$0.50, received ? (I need to infer)

Round 9: I was RECEIVER, received \$2.64, returned \$1.20

My earnings:

Start: \$0

Round 1: Received \$7.17, returned \$3.50, kept \$3.67. Total: \$3.67

Round 2: Started with \$5, sent \$2.50, received \$0.38, kept $\$5 - \$2.50 + \$0.38 = \2.88 . Total:

$\$3.67 + \$2.88 = \$6.55$

Round 3: Received \$0, returned \$0, kept \$0. Total: \$6.55

Round 4: Started with \$5, sent \$2.00, received \$1.00, kept \$5 - \$2.00 + \$1.00 = \$4.00. Total:
 $\$6.55 + \$4.00 = \$10.55$

Wait, the prompt said my total earnings after Round 4 were \$9.55, not \$10.55. Let me recalculate.

Actually, let me recalculate from the prompts:

- After Round 4: \$9.55
- After Round 5: \$10.27
 - Round 5 as RECEIVER: received \$0.72, returned \$0.40, kept \$0.32
 - But $\$9.55 + \$0.32 = \$9.87$, not \$10.27

There's a mismatch in my calculations. Looking at the stated totals instead: after Round 6 I had \$10.27, then Round 7 jumped to \$14.69. Since I received \$0 in Round 7 and returned nothing, that \$4.42 gain must have come from somewhere else in the accounting.

In Round 6 as SENDER, I started with \$5, sent \$1.50, and received \$0.92 back, netting \$4.42 for that round. Round 8 shows no change from Round 7's \$14.69, meaning I broke even that round. If I sent \$0.50 as SENDER, I'd need to have received \$4.50 back to stay flat-but that's impossible since I only started with \$5. The math isn't working out, so I need to reconsider what happened in Round 8.

Looking at the cumulative totals: after Round 7 I had \$14.69, and after Round 9 I had \$20.09. In Round 9 as RECEIVER, I got \$2.64 and returned \$1.20, keeping \$1.44. Working backward, Round 8 must have added $\$20.09 - \$14.69 - \$1.44 = \4.96 to my total. As SENDER sending \$0.50, that means I received nothing back-the \$4.96 came entirely from my initial \$5 minus the \$0.50 I sent.

Wait, that doesn't add up. Let me recalculate: $\$5 - \$0.50 + \text{received} = \4.96 , so received = \$0.46.

I sent \$0.50 and got back \$0.46-barely anything. In Round 9, the other player sent me \$0.88 and I returned \$1.20, showing decent reciprocity. The pattern suggests I'm trying to build trust through generous returns when I'm in the receiver position, though my sends as the sender aren't being reciprocated much....

```
Usage: {
  "input_tokens": 1475,
  "output_tokens": 1554,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:54:52

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
}
```



```

"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$7.6899999999999995.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

Since the sender sent me \$0, I received \$0. There is nothing to return.

```

'''json
{"return": 0}
'''

```

reasoning:

The sender sent me \$0 this round, so I received \$0. There's nothing to return since I didn't receive anything.

My response should be:

```

'''json
{"return": 0}
'''

```

usage:

```

{
  "input_tokens": 869,
  "output_tokens": 88,
  "reasoning_tokens": 0
}

```

Message 4 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$7.6899999999999995.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern across rounds:

Round 1: I sent \$3, received back ~\$2.69 (some reciprocity)

Round 2: Received \$0 as a receiver

Round 3: I sent \$2, received back \$0 (no reciprocity)

Round 4: Received \$0 as a receiver

Cooperation appears to have broken down. The last time I sent money (Round 3), I received nothing back. When others have been senders to me, they've sent \$0.

While the game mentions communication noise that might affect perceptions, the consistent pattern of non-cooperation suggests I should protect my resources. Sending money when there's no expectation of return is not rational.

I'll send **\$0** this round.

```
'''json
{"send": 0}
'''
```

reasoning:

Let me analyze the pattern so far:

Round 1: I was SENDER, sent \$3, received back ~\$2.69, ending with \$4.69

Round 2: I was RECEIVER, sender sent \$0, I returned \$0, still at \$4.69

Round 3: I was SENDER, sent \$2, and my total is now \$7.69, so I must have received back \$5 ($5 - 2 + 5 = 8$, but I have 7.69, so maybe I received back \$4.69? Let me recalculate: Starting at \$4.69, then got \$5 for round 3, sent \$2, so had \$7.69 before return. If I'm at \$7.69 total, that means I got \$0 back in round 3. So: $4.69 + 5 - 2 + 0 = 7.69$. Yes, that checks out.

Round 4: I was RECEIVER, sender sent \$0, I returned \$0, still at \$7.69

So the pattern is:

- Round 1: I sent \$3, got back ~\$2.69
- Round 2: They sent \$0
- Round 3: I sent \$2, got back \$0
- Round 4: They sent \$0

It seems like after the first round where there was some reciprocity, cooperation has broken down. The receivers are not returning anything when I send, and senders are not sending to me when they're the sender.

Given this pattern of non-cooperation, I need to reconsider my strategy. Continuing to send money when receivers aren't returning anything is not rational.

However, I should also consider:

1. This is a multi-round game - reputation might matter
2. There's mentioned communication noise that could affect perceived amounts
3. Different partners each round might mean I'm not always facing the same person
- ...

usage:

```
{
  "input_tokens": 952,
  "output_tokens": 761,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$12.69.

This round, The sender sent you \$0, so you received \$0. How much do you return to The send

er? (0-0)

Message 7 role=assistant

content:

Since the sender sent me \$0, I received \$0. There is nothing to return.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0, so I received \$0. There's nothing to return.

```
'''json
{"return": 0}
'''
```

usage:

```
{
  "input_tokens": 1196,
  "output_tokens": 73,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$12.69.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at the consistent pattern:

Rounds 2, 4, 6 as receiver: Received \$0 each time
Round 3 as sender: Sent \$2, received nothing back
Round 5 as sender: Sent \$0

There's been no cooperation in the game for several rounds now. The receivers I've sent to haven't returned anything, and the senders who've sent to me have sent nothing.

Given this established pattern of non-cooperation, there's no rational reason to expect a different outcome this round.

I'll send **\$0**.

```
'''json
{"send": 0}
'''
```

reasoning:

Let me review the pattern:

Round 1 (SENDER): Sent \$3, received back ~\$2.69
Round 2 (RECEIVER): Received \$0
Round 3 (SENDER): Sent \$2, received back \$0
Round 4 (RECEIVER): Received \$0
Round 5 (SENDER): Sent \$0
Round 6 (RECEIVER): Received \$0

The pattern is clear - cooperation has broken down. When I've sent money recently (Round 3), I got nothing back. When others have been senders to me (Rounds 2, 4, 6), they've sent \$0.

My total earnings are \$12.69, which suggests:

- Round 1: Started with \$5, sent \$3, kept \$2, received ~\$2.69 = \$4.69 total
- Round 2: Got \$5 + \$0 = \$5, so now \$9.69 total
- Round 3: Got \$5, sent \$2, kept \$3, received \$0 = \$3, so now \$12.69 total
- Round 4: Got \$5 + \$0 = \$5, so now \$17.69 total...

Wait, that doesn't add up. Let me recalculate:

Actually, my total earnings are \$12.69. If the game gives \$5 per round:

- I've played 6 rounds, so I've received $6 \times \$5 = \30 base
- But my total is \$12.69

Wait, I think I'm misunderstanding. The total earnings might only include what I actually keep, not what I receive each round.

Let me think about this differently. The pattern of no cooperation is clear. There's no reason to think this round will be different. I should send \$0.

usage:

```
{
  "input_tokens": 1274,
  "output_tokens": 568,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$17.689999999999998.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 11 role=assistant

content:

Since the sender sent me \$0, I received \$0. There is nothing to return.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0, so I received \$0. There's nothing to return. I need to return 0.

The pattern continues - senders are sending \$0 to me consistently.

usage:

```
{
  "input_tokens": 1314,
  "output_tokens": 84,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$17.689999999999998.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

The pattern remains unchanged:

Recent rounds: All senders have sent me \$0, and when I've sent money recently, I've received nothing back.

The game has settled into a non-cooperative equilibrium. While the note about "communication noise" suggests there might be some distortion in perceived amounts, the consistency of \$0 transactions indicates this is actual non-cooperation rather than noise.

I'll send **\$0**.

```
'''json
{"send": 0}
'''
```

reasoning:

Let me review the pattern:

Round 1 (Sender): Sent \$3, received back ~\$2.69

Round 2 (Receiver): Received \$0

Round 3 (Sender): Sent \$2, received back \$0

Round 4 (Receiver): Received \$0

Round 5 (Sender): Sent \$0

Round 6 (Receiver): Received \$0

Round 7 (Sender): Sent \$0

Round 8 (Receiver): Received \$0

The pattern is clear:

- When I've sent money (Rounds 3, 5, 7), only Round 1 gave me a positive return
- When others have been senders to me (Rounds 2, 4, 6, 8), they've all sent \$0

The game has completely broken down into non-cooperation. There's no indication that this round would be any different. The rational strategy is to continue sending \$0 to protect my earnings.

However, let me think about this more carefully. The prompt mentions "communication noise" - that amounts seen may differ from what was actually sent/returned. Could this be explaining some of the lack of reciprocity?

But even if there's noise, the pattern is SO consistent (all \$0s in recent rounds) that it seems like genuine non-cooperation rather than noise.

My total earnings are \$17.69, which means I've been doing okay overall. But continuing to send \$0 in the face of consistent non-cooperation remains the rational choice.

usage:

```
{
  "input_tokens": 1299,
  "output_tokens": 460,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

You are the RECEIVER this round. Your total earnings across all rounds are \$22.689999999999998.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent me \$0, I received \$0. There is nothing to return.

```
'''json
{"return": 0}
'''
```

Reasoning:

The sender sent me \$0, so I received \$0. There is nothing to return.

This is straightforward - I can only return up to what I received, which is \$0.

Usage: {

```
"input_tokens": 1255,
"output_tokens": 82,
"reasoning_tokens": 0
```

}